



AS chemistry

XCH11

Notes from MS
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Formula Sheet

for A&S Chemistry

Ideal Gas Equation

$$PV=nRT$$

$P \Rightarrow \text{Pa}$

$V \Rightarrow \text{m}^3$

$R \Rightarrow 8.31 \text{ kJ} \cdot \text{mol}^{-1}$

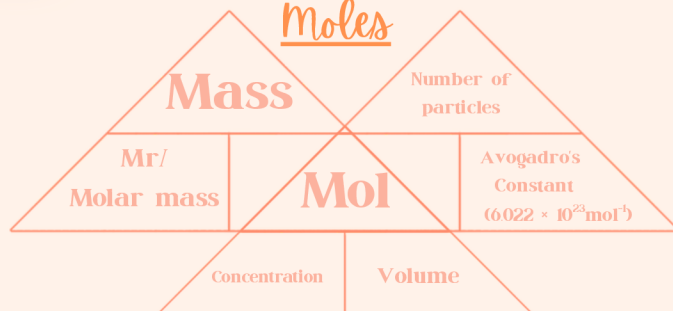
$n \Rightarrow \text{moles}$

$T \Rightarrow \text{K}$

* when moles are equal in a question:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Moles



$$\text{Mole (n)} = \frac{PV}{RT}$$

$$\text{Volume} = \text{mole} \times \frac{24(000)}{\text{dm}^3 \text{ cm}^3}$$

Ionisation Energy



$n \Rightarrow \text{degree of ionisation}$

(first, second,)

$X \Rightarrow \text{element}$

Relative Atomic Mass

$$\frac{\sum (\text{Isotopic masses} \times \% \text{ abundance})}{100}$$

$$\frac{\sum (\text{Isotopic masses} \times \text{relative abundance})}{\text{Total relative abundance}}$$

% of Atom Economy

Mass of desired product

Mass of all **Reactants** $\times 100$

* % = 100 when there's only one product

Diluting Concentrations

$$\frac{\text{Original Concentration} \times \text{Original Volume}}{\text{New diluted volume}}$$

* when moles are equal in a question: $\text{Orig. Vol.} \times \text{Orig. Conc.} = \text{New Vol.} \times \text{New Conc.}$

* New volume = Original + Volume of water added

% yield

$$\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

Parts per million

$$\frac{\text{Volume of Solute}}{\text{Volume of Solution}} \times 10^6$$

Density

$$\frac{\text{Mass}}{\text{Volume}}$$

Enthalpy and Energy Change

ΔH : $\sum \text{bond enthalpies broken}$

$- \sum \text{bond enthalpies formed}$

$$\frac{Q_{\text{exp}}}{\text{mol}}$$

Energy Change: $Q = MC\Delta T$

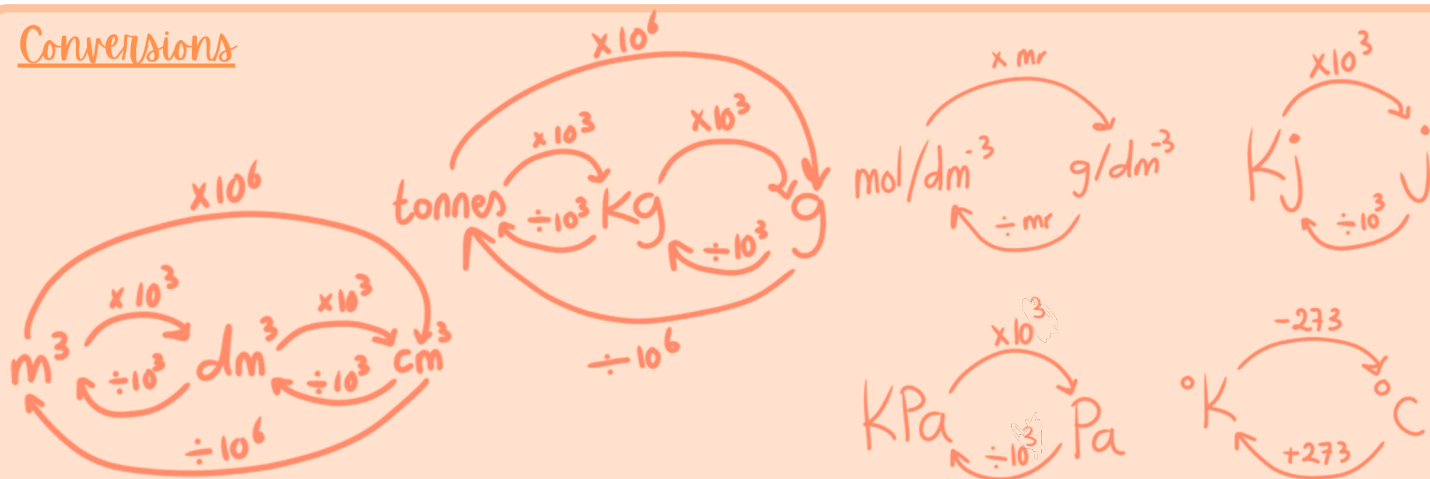
$M \Rightarrow \text{mass of Solution (g)}$

$C \Rightarrow \text{heat capacity (4.18 J g}^{-1} \text{K}^{-1})$

$\Delta T \Rightarrow \text{difference in temp. (K)}$

$Q \Rightarrow \text{energy change in (J)}$

Conversions



IAS Chemistry U2 Notes From MS

R.D.= Repeated Definition R.Q.= Repeated Question

Energetics

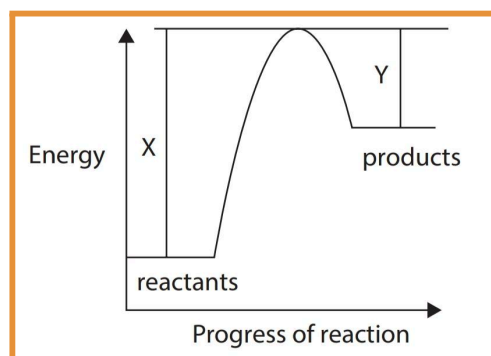
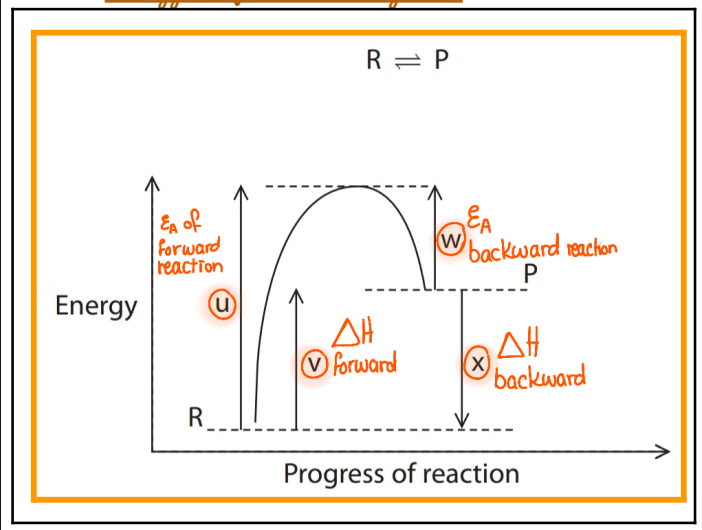
❖ Some Key Terms:

Term	Definition
Standard enthalpy change of formation	(the enthalpy change when) one mole of the substance (is formed) from its elements in their standard states (under standard conditions).
standard enthalpy change of neutralisation	heat energy released under standard conditions (when) 1 mol of water is produced (by the reaction of acid with alkali)
Mean bond enthalpy	energy required to break one mole of a (specific) bond in the gaseous phase/state averaged over a number of compounds/ different molecules/different compounds/different molecules.

❖ Classification of some ΔH :

$\Delta_{at}H$	$\Delta_c H$	$\Delta_{neut}H$	$\Delta_f H$
always positive (endothermic)	always negative (exothermic)		either positive or negative

❖ Energy Profile/Level Diagrams:



X is E_a of forward reaction, Y is E_a of backward reaction, $X+Y$ is the sum of all E_a , $X-Y$ is the enthalpy change.

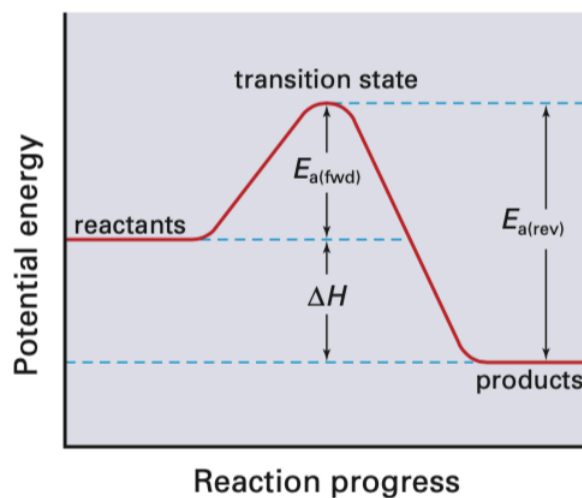


Figure 6.12 A potential energy diagram for an exothermic reaction

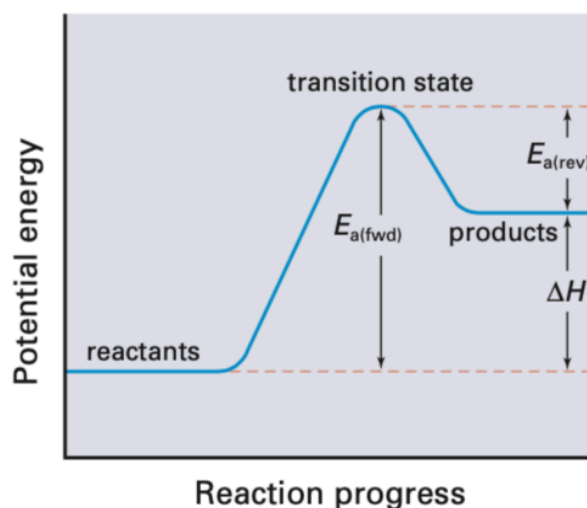


Figure 6.13 A potential energy diagram for an endothermic reaction

❖ Formulas:

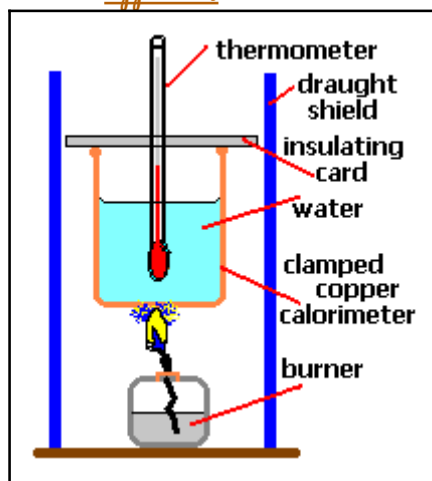
$$Q = mc\Delta T$$

HEAT ENERGY (J) MASS OF WATER (g) SPECIFIC HEAT CAPACITY TEMPERATURE CHANGE

$$\Delta H = \frac{Q}{\text{moles}}$$

HEAT ENERGY (J) ENTHALPY CHANGE (J/mol)

ENTHALPY CHANGE (ΔH) = ENERGY TO BREAK BONDS – ENERGY RELEASED MAKING BONDS

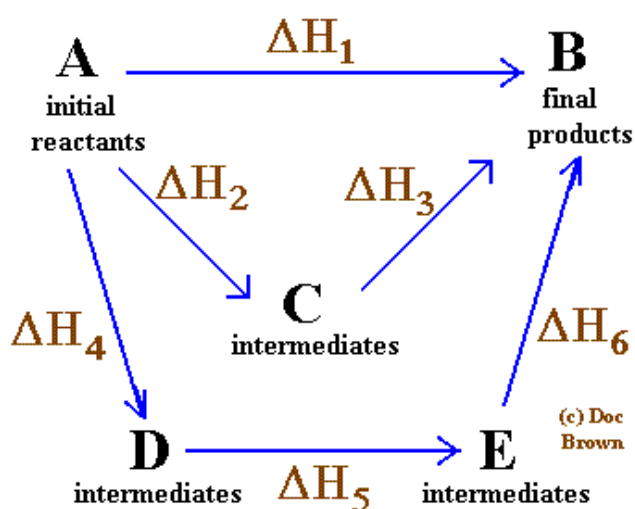
❖ Combustion of Alcohols:> Apparatus> Comparing results to Data Booklet:

(R.Q)
Obtained
value
< D.B.
(Inaccurate
value)

1. *heat loss* from the (copper) calorimeter to *surroundings*.
2. *incomplete combustion* of the alcohol.
3. *loss* of alcohol by *evaporation*/some of the alcohol *evaporates*.
4. calculation *does not take into account* the *heat capacity* of the *beaker*.
5. use of *non-standard conditions* (*least likely*).
6. (If mean bond enthalpies are used for the calculation) :(*mean*) bond enthalpies *do not refer to specific compounds* OR uses *mean bond enthalpies* (rather than *for specific compounds*) OR *mean bond enthalpies* are *averages/ mean for different molecules/bonds in different environments/compounds* OR bond enthalpies are *averaged over a (large) number of compounds*.

7. (If a liquid alcohol is used): [named] *alcohol* is a *liquid* and *bond enthalpies* refer to *gases/gaseous state* OR *mean bond enthalpy* calculations *do not include changes of state*.

- Obtained value > D.B.
1. *use of the molar mass of alcohol in the calculation*.

❖ Hess's Law & Cycle:

Enthalpy change of reaction from enthalpy changes of formation

We can calculate the enthalpy change of reaction by using the type of enthalpy cycle shown in Figure 6.8.

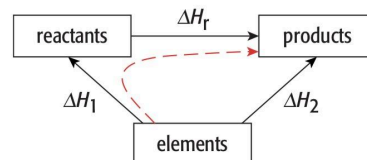


Figure 6.8 An enthalpy cycle for calculating an enthalpy change of reaction. The dashed line shows the indirect (two-step) route.

Enthalpy change of formation from enthalpy changes of combustion

We can calculate the enthalpy change of formation of many compounds by using the type of enthalpy cycle shown in Figure 6.10.

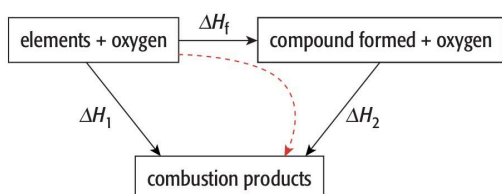


Figure 6.10 An enthalpy cycle for calculating an enthalpy change of formation from enthalpy changes of combustion. The dashed line shows the two-step route.

Calculating the enthalpy change of hydration of an anhydrous salt

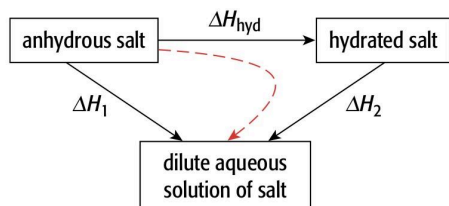
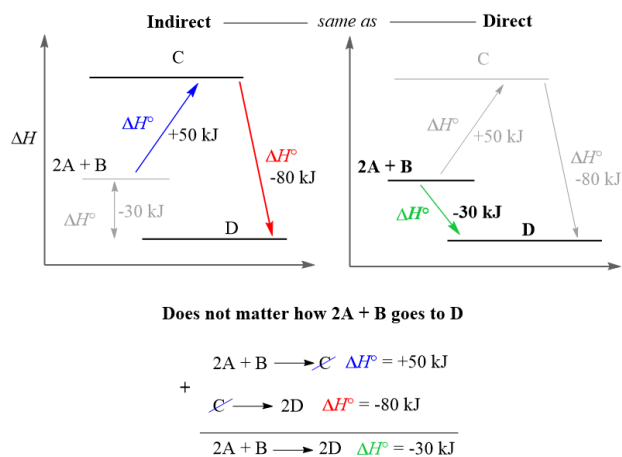


Figure 6.12 An enthalpy cycle to calculate the enthalpy change of hydration of an anhydrous salt. The dashed line shows the two-step route.

Hess's Law

Because enthalpy is a state function, it does not matter how "2A + B" goes to "2D". Therefore, we add the enthalpy changes for the two reference reactions.

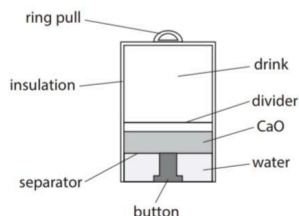


★ Exam Questions:

- A graph of the enthalpy change of combustion against the number of carbon atoms is approximately a straight line from ethane to hexane. Explain this statement by considering the bonds broken and made in the combustion of alkanes.
 - (because) for each increase of carbon chain by 1 there is 1 more C–C bond and 2 more C–H bonds broken (and 1½ O=O bonds broken) and two more C=O and two more O–H bonds made.
- Give a reason why it is not possible to measure the enthalpy change for the decomposition of magnesium carbonate directly.
 - it is difficult to measure the temperature of a solid/powder/ while it is being heated.
- Give a reason why it is not possible to measure the enthalpy change for the decomposition of Sodium hydrogencarbonate directly.
 - hard to measure the temperature change when you're heating something OR heat losses due to high temperatures involved OR at 300C/high temperatures the water will be gaseous.
- (R.Q.): State why the standard enthalpy change of formation of oxygen is 0.0 kJ mol⁻¹.
- State why no Δ_fH value has been given for oxygen.
 - oxygen/it is an element in its standard state.
- Explain why an Δ_fH can also be described as an enthalpy change of formation.
 - 1 mol of substance / compound formed from element(s) in standard state(s).
- Explain how, if at all, the enthalpy change of neutralisation obtained would differ if the heat capacity of the calorimeter was included in the calculation.
 - (because the calculation has not taken into account the) energy required to heat the calorimeter/ the (total) heat capacity would be greater, so the value(of the enthalpy change of

neutralisation) would be more exothermic/more negative.

The reaction between calcium oxide and water to produce calcium hydroxide solution is exothermic, and is used to warm canned drinks. A typical arrangement is shown.

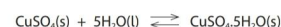


- Suggest two reasons why the can is insulated.
 - (reason 1) to keep the drink at the required temperature / to minimise heat loss / reduce energy transfer to surroundings
 - (reason 2) to allow the can to be handled safely
 - (reason 3) high thermal conductivity of the can
- Give two reasons why the energy used to move the car is less than the theoretical maximum.
 - heat loss to the surroundings
 - energy is used to bring the engine to operating temperature
 - incomplete combustion (of the fuel)
- In the calculation, the specific heat capacity of the liquid is taken to be $4.18 \text{ Jg}^{-1} \text{ } ^\circ\text{C}^{-1}$ rather than the true value of $3.85 \text{ Jg}^{-1} \text{ } ^\circ\text{C}^{-1}$. The use of $4.18 \text{ Jg}^{-1} \text{ } ^\circ\text{C}^{-1}$ in the calculation lowers the error in the final value obtained.

- (R.Q.): The equation for standard enthalpy change of atomisation, $\Delta_{\text{at}}H$, of bromine:
 - $(\frac{1}{2}\text{Br}_2(\text{l}) \rightarrow \text{Br}(\text{g}))$
- State the **two** standard conditions for enthalpy change.
 - (temperature) 298 K / 25°C and (pressure) 1 atm / 100 kPa / 101 kPa / $1 \times 10^5 \text{ Pa}$ / $1.01 \times 10^5 \text{ Pa}$.
- The equation for a reaction is $2\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g})$ Which is the correct symbol for the enthalpy change for this reaction?
 - $(\Delta_r H)$
- The standard enthalpy change of neutralisation for the reaction between sodium hydroxide solution and hydrochloric acid is -56 kJmol^{-1} .
 - The reaction is exothermic and the temperature increases.

- A student carried out experiments to determine the value of the enthalpy change, $\Delta_r H$. Known masses of anhydrous and hydrated copper(II) sulfate were dissolved separately in water in insulated containers, and the temperature changes measured. State why different volumes of water are used in the two experiments.
 - the hydrated crystals already have a certain amount/ 4.5 g / 5 moles of / (more) water
OR 4.5 g is the difference in mass between the hydrated and anhydrous salt.
 - so that the total amount/mass/ volume of water is kept the same/ 50 cm^3 / 58 g.

(b) The hydration of anhydrous copper(II) sulfate is reversible.



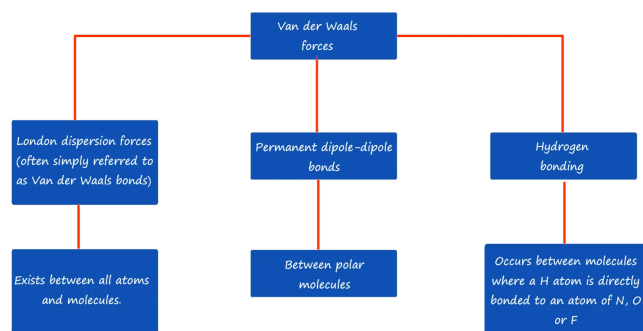
The forward reaction is exothermic. The temperature changes for both the forward and reverse reactions are difficult to measure.

- Suggest a reason in each case.
 - Forward
 - adding water to the anhydrous salt will produce a solution and does not form a solid
 - some of the water may turn to steam/be lost (because the forward reaction is exothermic)
 - difficult to determine when exactly 5 mol water to 1 mol CuSO_4 has been added to form a solid
 - the temperature of a solid is difficult to measure
 - Reverse
 - the hydrated salt has to be heated/ the energy from the Bunsen burner is difficult to take into account/ hard to measure the temperature when heating
 - the temperature of a solid is difficult to measure
 - the salt may decompose further.
- Explain why the increases in the values of $\Delta_c H$ are similar in alkanes.
 - each alkane differs from the next by $-\text{CH}_2$ /one carbon atom and two

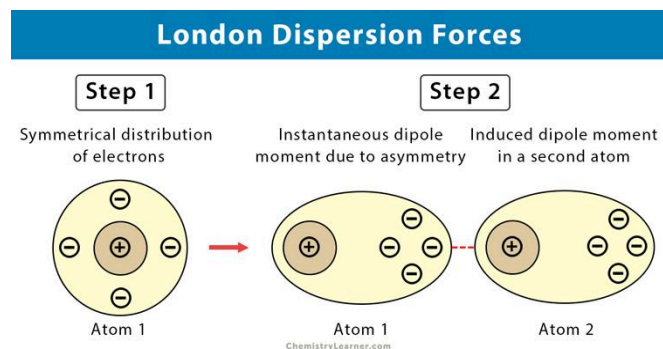
hydrogen atoms/same number of C-C bonds and same number of C-H bonds/ forms a homologous series/ has a general formula C_nH_{2n+2}

- bonds broken and/or made increase regularly/ each $-CH_2$ combusted adds almost the same amount of energy/ bond energies are very similar.
- The increase in the value of $\Delta_c H$ from butane to pentane is smaller than any other increase. Suggest an explanation for this.
 - butane is a gas and pentane is a liquid (at 298 K / under standard conditions)
 - liquids vaporise before burning/combustion
 - (pentane is a liquid so) some of the energy released by combustion is used to vaporise / vapourisation is endothermic OR more energy is needed to break intermolecular forces in pentane or reverse argument.

❖ IMF:

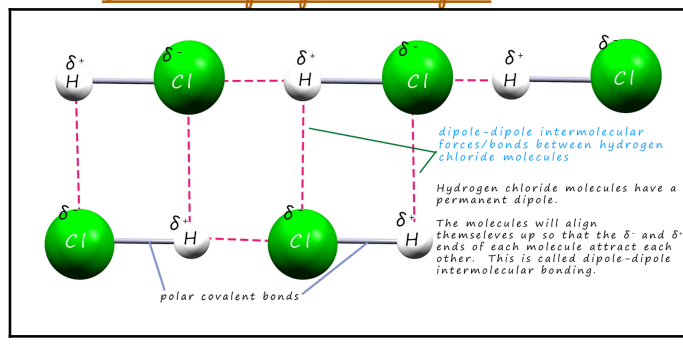


❖ London forces (dispersion forces/instantaneous-induced dipole):

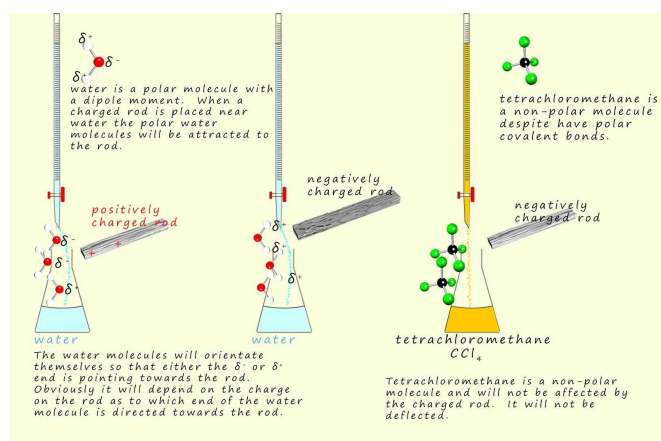


- Describe how London forces arise.
 - instantaneous/temporary dipole formed by asymmetrical distribution of charge/ random arrangement of electrons, these produce induced dipoles.
- Strongest LFs = highest number of electrons.
- State why the London forces in hydrogen iodide are stronger than those in hydrogen chloride and hydrogen bromide.
 - (hydrogen iodide has) more electrons.

❖ Permanent dipole-permanent dipole:



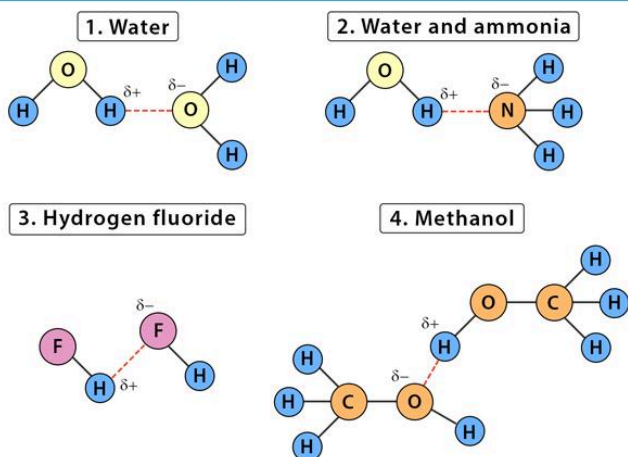
Intermolecular Forces



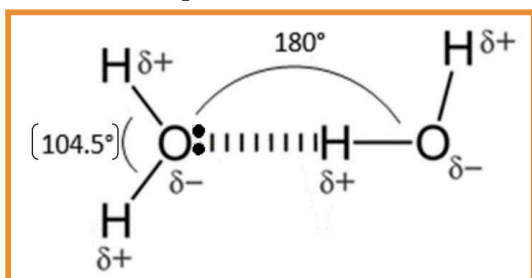
- Silane, SiH_4 , does not contain permanent dipole.

❖ Hydrogen bonding:

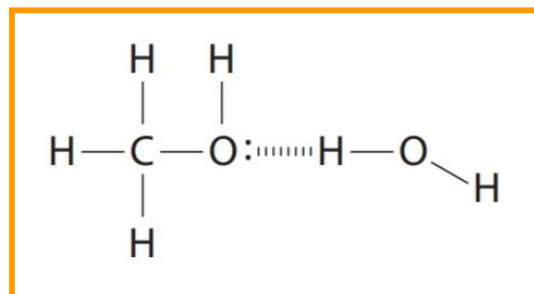
Hydrogen Bond Examples



- State the reason why hydrogen bonds do not form between ethanal molecules.
 - none of the hydrogen atoms in ethanal are bonded to an oxygen atom (or another highly electronegative atom)*
- Draw a diagram to show hydrogen bonding between two H_2O molecules in ice.



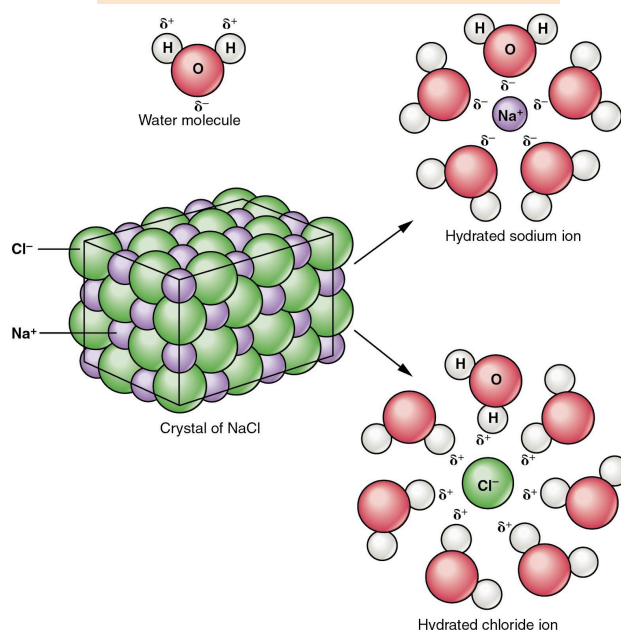
- Which structure best represents a hydrogen bond between methanol and water?



- The more hydrogen bonds formed/the more OH gps in a compound, the more viscous it is.
- hydrogen bonding explains why ice has a lower density than water, HF has a higher boiling temperature than HCl, and alcohols are less volatile than similar alkanes.

❖ Ion-permanent dipole:

- (R.Q.): The attraction/interaction forces between ions and water is ion-permanent dipole.



- Describe the processes that occur when solid copper(II) sulfate dissolves in water.
 - (water molecules) break the (ionic) lattice/solid and can hydrate/surround the ions / ion-dipole interactions form.*

• Properties of Compounds due to IMFs:

Comparing Boiling Temperatures

London Forces

- **Weakest** IMF, takes **little** energy to break, so **lowest** boiling temperature.
- If two or more molecules have the **same number of electrons**, then they have the **same London Forces**, so cannot account for any difference in b.p.

PD-PD interactions

- **Stronger** than LFs, but **weaker** than Hydrogen Bonds.
- LFs + Permanent Dipoles = **higher** temperature (than LFs alone)

Hydrogen bonds

- **Strongest** IMFs, takes **more** energy to break, so **highest** boiling temperature.
- In terms of **strongest hydrogen bond** ($H \cdots M$): $H \cdots F > H \cdots O > H \cdots N$.
- The **more hydrogen bonds** in a compound, the **more energy required** to break, thus the **higher** the **boiling point**.
- LFs + Permanent dipoles + H bonds = **highest** boiling point.

Comparing between molecules' b.p.s.

1. **State the IMFs in all the molecules and explain why they have them:**
 - a. Only forms **LFs**? **Non-polar, symmetrical** molecule.
 - b. Forms both LFs and **PD-PD interactions**? **Polar** molecule with a polar bond.
 - c. Forms LFs, PD-PD interactions, and **Hydrogen Bonds**? Contains **hydrogen** attached to a **small electronegative element (F, O, N)**
2. If molecules are compounds of elements down a specific group (e.g. group 5 hydrides):
 - a. **number of electrons increases down a group** so London dispersion forces become **stronger**.
3. If molecules both have hydrogen bonds:
 - a. **Compare number of hydrogen bonds per molecule**. The higher the number, the higher the b.p.
4. Finally, state that **hydrogen bonds** (if available in any of the molecules being compared) are the **strongest intermolecular forces** so take the most/more energy to break, and **London**

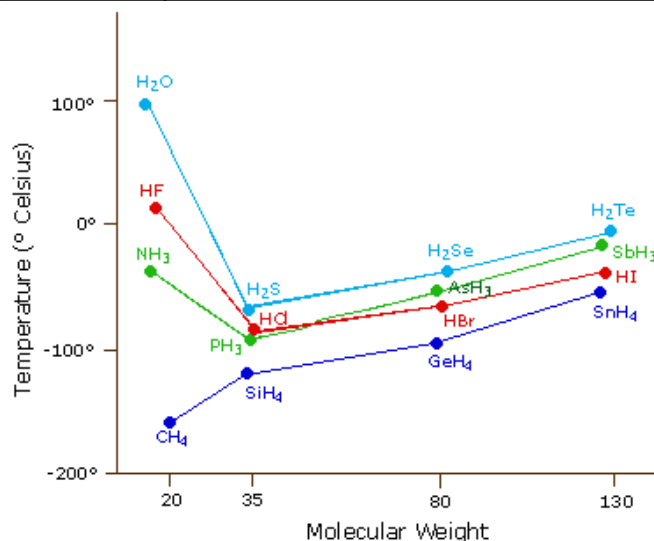
forces are the **weakest intermolecular forces**, so molecule has the **lowest boiling temperature**.

Two or more molecules, different IMFs & no. of e^s but similar b.p.s.

1. **State the strongest IMFs** in those compounds.
2. the intermolecular forces in the two compounds are **similar in strength** / require a **similar amount of energy** to break.

Two or more molecules of same type, and same IMFs, but different b.p.s

- If both have LFs:
 1. **stronger London forces** (as) **greater (contact) surface area** (between molecules) [in **organic compounds**]
 - ★ the **more branching** in an organic compound = the **smaller the surface area**, the **weaker the LFs**, and the **lower the boiling point**.
 2. **Stronger LFs** due to having **more electrons**.
- If both have permanent dipole-permanent dipole interactions:
 1. **stronger PD-PD interactions** due to having a **more polar bond** (a bond with **greater difference in electronegativity**).



Trend in boiling points of hydrides

- State what causes hydrogen fluoride to have such an unusually high boiling temperature.

- hydrogen fluoride has hydrogen bonding (the strongest intermolecular force).
- Explain the trend in the boiling temperatures of the hydrogen halides.
 - all hydrogen halides have London forces (and dipole-dipole forces between molecules)
 - the strength of the London forces increases as the number of electrons increases (so the boiling temperature increases from HCl to HI) OR the strength of the London forces increases as the polarisability of the molecules increases from HCl to HI
 - (only) HF has hydrogen bonding (between molecules)
 - hydrogen bonding is (much) stronger than London forces / dipole-dipole forces (so HF has the highest boiling temperature)
- Explain, using intermolecular forces, why iodine monochloride has a very similar boiling temperature to water, despite having many more electrons.
 - iodine monochloride/ICl has London forces as the strongest intermolecular force, while water/H₂O has hydrogen bonds/H-bonds (as the strongest intermolecular force).
 - the intermolecular forces (in ICl and H₂O) are similar in strength / require a similar amount of energy to break.

Hydride	Boiling temperature / K
NH ₃	240
PH ₃	186
AsH ₃	212
SbH ₃	256

Explain the differences in the boiling temperatures of the Group 5 hydrides shown.

- the hydrides have London forces

- number of electrons increases so London dispersion forces become stronger (down the group)
- (only) ammonia has hydrogen bonding, (which) is (much) stronger than the London forces (in ammonia)

(b) Compare and contrast the effect of intermolecular forces on the properties of water and ammonia, using the data shown.

(3)

Molecule	M_r	Boiling temperature / K	Density at 223 K / kg m ⁻³	Density at 278 K / kg m ⁻³
NH ₃	16.0	240	698	0.763
H ₂ O	18.0	373	926	1000

- molecules are a similar size / same number of electrons so the London forces are similar (and cannot account for big difference in boiling temperature)
- there are more **hydrogen bonds** between water molecules (than hydrogen bonds between ammonia molecules, resulting in water having a higher boiling temperature than ammonia)
- density of ammonia decreases between the two temperatures as it turns (from a liquid) to a gas OR density of water increases between the two temperatures as it turns (from a solid) to a liquid.
- *Two alternatives to hydrocarbon fuels are hydrogen and ammonia. Hydrogen burns to form water as the only product but hydrogen is extremely hard to liquefy. Ammonia is easily liquefied but it is toxic. Compare and contrast the intermolecular forces involved when hydrogen and ammonia liquefy, by describing how these forces are formed.
 - **Similarities**
 - both hydrogen and ammonia form London / dispersion forces
 - a temporary dipole forms in a molecule and induces a dipole in an adjacent molecule
 - the attraction (between the temporary dipoles) is small(er) in hydrogen because the H₂ electron cloud is not easily polarised
 - **Differences**

- ammonia forms hydrogen bonds (because nitrogen is very electronegative)
 - Accept hydrogen bond forms between the nitrogen lone pair and the (δ^+) hydrogen (of a different molecule)
 - ammonia liquefies more easily than hydrogen because hydrogen bonds are stronger than London forces.
- Commercial disinfectants are often made more viscous by adding propane-1,2,3-triol to the aqueous solution. Suggest how propane-1,2,3-triol makes disinfectants more viscous.
 - propane-1,2,3-triol forms three hydrogen bonds per molecule
 - so the molecules are strongly attracted to each other (and to surfaces)
- Explain why alcohol B has a higher boiling temperature than alcohol A.
 - (alcohol B has) stronger London forces (as) greater (contact) surface area (between molecules).
- Urea has a melting temperature of 133°C . Explain why this value is higher than expected for a relatively small molecule.
 - urea forms hydrogen bonds and (permanent) dipole-dipole forces.
 - these forces are stronger than London forces

Some data for three small molecules are shown.

Molecule	M_r	Boiling temperature /K
Fluorine	38.0	85
Hydrogen sulfide	34.1	212
Methanol	32.0	338

Some data for three small molecules are shown.

Molecule	M_r	Boiling temperature / $^\circ\text{C}$
Fluorine	38.0	-188
Hydrogen chloride	36.5	-85
Methanol	32.0	65

(R.Q.): *Explain the large variation in boiling temperatures, given the small range in M_r values.

- fluorine molecules only have London forces (instantaneous dipole-induced dipole) between them (as it has a symmetrical electron cloud/is a symmetrical/non-polar molecule.)
- hydrogen chloride/hydrogen sulfide is a polar molecule as chlorine/sulfur is more electronegative than hydrogen / H-Cl/H-S bond is polar, so $\text{HCl/H}_2\text{S}$ forms permanent dipole-permanent dipole interactions in addition to London forces.
- methanol contains a hydrogen attached to a small electronegative element so can form hydrogen bonds (in addition to permanent dipole-permanent dipole interactions and London Forces).
- hydrogen bonds are the **strongest** intermolecular forces so take the most/more energy to break.
- London forces are the **weakest** intermolecular forces, so fluorine has the lowest boiling temperature.

This question is about trends in the Periodic Table.

*(a) The boiling temperatures of some isoelectronic hydrides are shown.

Hydride	CH_4	NH_3	H_2O	HF
Boiling temperature / K	112	240	373	293

The boiling temperatures of some isoelectronic hydrides are shown.

Hydride	CH_4	NH_3	H_2O	HF
Boiling temperature / $^\circ\text{C}$	-161	-33	100	20

(R.Q.): *Explain the differences in these boiling temperatures by considering all the intermolecular forces involved.

- methane has (only) London forces / dispersion forces / instantaneous dipole - induced dipole forces, and London forces are weaker than hydrogen bonds, so boiling temperature of methane is the lowest.
- $\text{NH}_3/\text{H}_2\text{O}/\text{HF}$ have hydrogen bonds, and London forces are same/similar as they have the same number of electrons
- the hydrogen bond is stronger in HF than $\text{H}_2\text{O}/\text{NH}_3$, as fluorine has higher electronegativity than N or O / is the most electronegative element/ H-F

bond is the most polar, so HF has a higher boiling point than ammonia.

- *Water has more hydrogen bonds (than ammonia/ HF) (so boiling point of water is the highest).*

- At room temperature and pressure, sulfur is a solid and oxygen a gas. Explain why sulfur and oxygen exist in different states by referring to the intermolecular forces involved.

- *both have London forces (only)*
- *S molecules have more electrons / S is a larger molecule (than oxygen) / S electrons are more easily polarised*

- The main reason that cyclohexane has a higher boiling temperature is that cyclohexane molecules (cyclohexane molecules have a larger area of contact).

The boiling temperatures of some halogenoalkanes are shown.

Halogenoalkane	Boiling temperature / °C
1-chloropropane	47
1-bromopropane	71
1-iodopropane	103

- Explain the trend in boiling temperature of these halogenoalkanes by comparing the intermolecular forces involved.
 - *London forces (and (permanent) dipole-(permanent) dipole forces)*
 - *iodine atoms are more polarisable (than chlorine or bromine)/ 1-iodopropane has more electrons (than 1-chloropropane and 1-bromopropane)*
 - *(resulting in) stronger / more London Forces (so more energy required to overcome these forces)*
 - *(despite) 1-iodopropane having the weakest (permanent) dipole / (permanent) dipole forces.*
- Explain why bromine is a liquid but iodine is a solid at room temperature.
 - *Iodine has more electrons (per molecule) (than bromine), so stronger London forces between molecules/iodine (mean a higher melting temperature for iodine)*
- Explain, with reference to their intermolecular forces, why the boiling temperatures of **alkanes**

increase as the number of carbon atoms increases.

- *the intermolecular forces are London / dispersion forces / instantaneous dipole – induced dipole forces*
- *these forces depend on the number of electrons which increase (as the number of carbon atoms increase)*
- *more **energy** is needed to break these intermolecular forces/to separate the molecules (so the boiling temperature increases)*
- Explain the difference in boiling temperature between butan-1-ol and 2-methylpropan-2-ol.
 - *(butan-1-ol has a higher boiling temperature than 2-methylpropan-2-ol because)*
 - *the instantaneous dipoles-induced dipoles / London forces / dispersion forces / van der Waals forces are stronger between straight chains*
 - *the straight chain molecule/ butan-1-ol has greater surface area / more points of contact.*

❖ Solubility:

Comparing solubilities

Solubility in water

- *[first bullet point]: water molecules form hydrogen bonds (and London forces and permanent dipoles), and hydrogen bonds are stronger than London forces/ are the strongest IMF.*
- *If completely or mostly soluble:*
 - *(first bullet point) + when the compound mixes with water, both can form hydrogen bonds (to each other)/ molecule can form hydrogen bonds to water, as the molecule and water have hydrogen bonds between their molecules. So, forces that form are similar in strength or stronger than hydrogen bonds (broken) in water (/ compound), so the compound dissolves.*

- If only slightly soluble (and molecule has $-OH$, $-NH$, $-NH_2$, or $-COOH$ (polar) groups):
 - (first bullet point) + some hydrogen bonds formed between molecule and water, aiding slight solubility, but, the intermolecular forces (formed) between the molecule and water are weaker than those in the molecule and/or in water, so cannot fully dissolve.
- If only slightly soluble or insoluble (and no polar gps present in the molecule):
 - (first bullet point) + molecule cannot form hydrogen bonds/ only forms weak London forces with water, so the (hydrogen) bonds between water molecules cannot be broken (so molecule does not dissolve in water).
- If the compound is ionic, and is insoluble:
 - large amount of energy required to break ionic bonds (in lattice / solid), but small amount of energy released during hydration (of ions) / when ions form bonds to water OR lattice energy is more exothermic than the hydration enthalpies.

Solubility in non-polar solvents (e.g. hexane) OR solvents of the same type as molecule (e.g. alcohol dissolving in ethanol solvent)

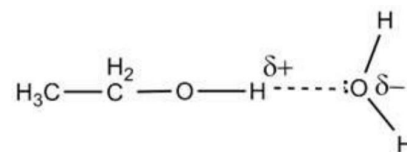
- molecule has only LFs, but cannot form hydrogen bonds. Also, the solvent forms only LFs, but cannot form hydrogen bonds.
- London forces formed between molecule and solvent are similar (in strength) to those (broken) in solvent and/or in molecule (so molecule dissolves/ thus, aiding (complete) solubility of molecule).

(d) Data about some physical properties of propan-2-ol and propanone are shown.

Compound	Boiling temperature / °C	Solubility in water
propan-2-ol	82.6	fully miscible
propanone	56.1	fully miscible

Compare and contrast these physical properties. Justify your answer in terms of intermolecular forces.

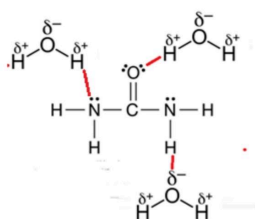
- (both soluble in water) as they can both form hydrogen bonds to water.
- propan-2-ol (has a higher boiling temperature than propanone, as it) forms hydrogen bonds, but propanone (only) forms (London forces and) (permanent) dipole-dipole attractions.
- hydrogen bonds are the strongest intermolecular force / are stronger than dipole-dipole attractions, therefore require more energy to break.
- Give a possible reason why ethanol is able to absorb water.
 - ethanol forms hydrogen bonds (with water).
- Explain why ethanol and water mix together fully.



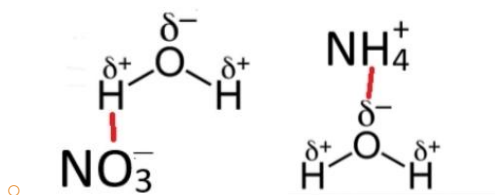
- when they mix can form hydrogen bonds (to each other)
- as both compounds have hydrogen bonds (between their molecules) OR forces that form are similar in strength or stronger than hydrogen bonds in water / ethanol
- from the lone pair / slight negative charge on an oxygen (atom in one molecule) to a slightly positive hydrogen (atom in the other molecule) (on OH group or water).
- *Explain why iodine is more soluble in hexane than in water, by considering the intermolecular forces in iodine, hexane and water and any intermolecular forces formed between iodine and the solvents.
 - iodine has (only) London forces/cannot form hydrogen bonds
 - water molecules form hydrogen bonds (and London forces and permanent dipoles)

- hydrogen bonds are stronger than London forces/ the strongest (intermolecular force)
- hexane forms (only) London forces/cannot form hydrogen bonds
- London forces formed between iodine and hexane are similar (in strength) to those (broken) in hexane/ iodine (so iodine dissolves)
- Iodine cannot form hydrogen bonds/ only forms weak London forces with water so the (hydrogen) bonds between water molecules cannot be broken (so iodine does not dissolve in water).
- *Both urea and ammonium nitrate are soluble in water. Discuss the differences in the interactions of water molecules with both urea and ammonium nitrate.

- **hydrogen bonds** are formed between water and urea
- urea is soluble in water as the hydrogen bonds formed are stronger/ similar in magnitude to the intermolecular forces between the individual molecules.



- ammonium nitrate is soluble in water as the water **hydrates** the ions (and the bonds formed are stronger/ similar in magnitude to the ionic bonds)

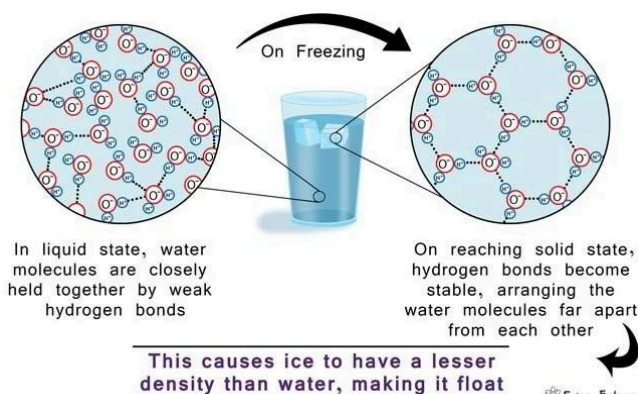


- Explain why alcohol B is completely soluble in ethanol but only slightly soluble in water.
 - M1 – London forces
 - London forces between B and ethanol (aiding complete solubility)
 - M2 – hydrogen bonds

- hydrogen bonds between B and water (aiding slight solubility)
- M3 – comparison of intermolecular forces formed and broken
 - intermolecular forces (formed) between B and ethanol are stronger than / similar in strength to those in B and/or in ethanol OR intermolecular forces (formed) between B and water are weaker than those in B and/or in water.
- Explain why iodine is very soluble in cyclohexane but only slightly soluble in water.
 - Cyclohexane and iodine form London forces (between molecules)
 - Hydrogen bonds between water molecules are stronger than London forces (between iodine and water molecules so less soluble in aqueous layer).
- Explain why petrol is more miscible with biobutanol than with bioethanol.
 - biobutanol has a longer hydrocarbon / alkane chain/more electrons than bioethanol
 - so more/stronger London forces / dispersion forces / Van der Waals forces between biobutanol and petrol (than bioethanol and petrol).
- Solubility of polar compounds is lowest in non-polar solvents.
- Which solvent dissolves the greatest amount of hydrocarbon C₃₅H₇₂? hexane.
- Explain, in terms of energy changes, why magnesium carbonate is insoluble in water.
 - large(r) amount of energy required to break **ionic** bonds (in lattice / MgCO₃ / solid), but small(er) amount of energy released during hydration (of ions) / when ions form bonds to water.
 - OR Lattice energy is **more exothermic** than the hydration enthalpies.

❖ Ice & Water:

WHY DOES ICE FLOAT ON WATER



- Water forms **two** or **four** hydrogen bonds per molecule, thus:
 - more open structure of ice.**
 - reduction in density of ice.**
 - expansion of volume of ice.**
 all compared to water.
- the H-O-H bond angle is **not** the same in ice and in water.

Redox

❖ Oxidation Number/State Rules:

Elemental state and Compounds	O.N. = ALWAYS 0 → The sum of O.N. is always equal to 0
Ions (monoatomic and polyatomic)	O.N. = the charge on the ion → Sum of O.N. = charge on ion
Group 1	O.N. = ALWAYS +1
Group 2	O.N. = ALWAYS +2
Hydrogen	O.N. = +1 when bonded to a

	non-metal O.N. = -1 when bonded to a metal (metal hydrides)
Oxygen	- O.N. = -2 (mainly) - O.N. = -1 in peroxides
Fluorine	O.N. = ALWAYS -1

❖ Oxidation, Reduction, Oxidising Agent and Reducing Agent:

Oxidation [Reducing agent = is oxidized but reduces another substance]	→ $X \rightarrow X^{+} + e^{-}$ → In terms of electrons: ◆ Loss of electrons, but an increase in Oxidation Number (e.g. from -m to +n, from 0 to +n, from +z to +n (where $n > m$ and z), from -m to -l (where $l < m$)) → In terms of H and O: ◆ Gaining of oxygen, but loss of hydrogen.
Reduction [Oxidising Agent = is reduced but oxidises another substance]	→ $Y^{+} + n e^{-} \rightarrow Y$ → In terms of electrons: ◆ Gaining of electrons, but a decrease in Oxidation Number (e.g. from +n to -m, from +n to 0, from +n to +z (where $n > m$ and z), from -l to -m (where $l < m$)) → In terms of H and O: ◆ Gaining of hydrogen, but loss of oxygen.

- State why the reaction between potassium chloride and conc sulfuric acid is not a redox reaction.
- State why Reaction 1 is not a redox reaction.
 - **no oxidation number has changed (between reactants and products)**

Ozone can be used in the treatment of drinking water. As well as killing bacteria and viruses, ozone also removes other dissolved impurities such as hydrogen sulfide, H_2S . Hydrogen sulfide reacts with ozone to produce sulfuric acid.



- State the role of the ozone in this redox reaction. Justify your answer using oxidation numbers.
 - (ozone is an) oxidising agent/oxidant

- S goes from -2 (in H_2S) to $+6$ (in sulfuric acid) **and** is oxidised
 - O goes from 0 (in ozone) to -2 (in sulfuric acid) **and** is reduced.
- Explain, in terms of oxidation numbers, why the reaction between calcium and chloride is a redox reaction.
 - calcium/Ca oxidised **and** from 0 (in Ca) to $(+2)$ (in $CaCl_2$)
 - chlorine/ Cl_2 /Cl reduced **and** from 0 (in Cl_2) to -1 (in $CaCl_2$).
- State which element is oxidised and which is reduced in the reaction between potassium and cold water.
 - K / potassium from 0 to $+1$ and oxidation
 - H / hydrogen from $+1$ to 0 and reduction



- (R.Q.): hydrogen **ions** act as **oxidising** agents.
- In which pair of compounds is the stated element in the **same** oxidation state? (**chromium in $K_2Cr_2O_7$ and K_2CrO_4**)
- Hydrogen peroxide, H_2O_2 , breaks down into water and oxygen. In terms of oxidation and reduction, how do hydrogen and oxygen change in this reaction? (**Hydrogen unchanged, Oxygen oxidised and reduced**)
- The oxidation number of sulfur in the sulfate ion, SO_4^{2-} , is **$+6$**
- Oxidising agents: **gain electrons, are reduced, decreased oxidation number.**

❖ Balancing & Combining half equations:

M	→ Balance the Main element of the equation: ◆ e.g.: $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$
O	→ Balance the Oxygen atoms by adding water: ◆ e.g.: $Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 7H_2O$ ★ If the product has oxygen but not the reactant, add the water in the reactant's side (LHS) and vice versa.
H	→ Balance the Hydrogen by adding H^+: ◆ e.g.: $Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$ ★ If the product has oxygen but not the reactant, add

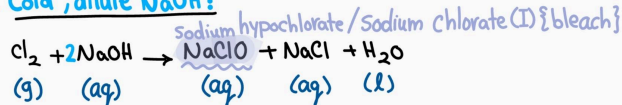
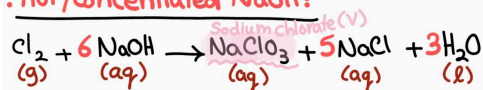
the water in the reactant's side (LHS) and vice versa.

E	→ Balance the Electrons by: [Total number of charge in reactants - total number of charge in products]. ◆ e.g.: $Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$ $[-2 + (14 \times 1)] - [(2 \times 3) + 0] = 6$ <u>Final equation:</u> $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
S	→ Write the state symbols if necessary or asked for in the exam question.
M	→ Repeat process in the other half equation: ◆ e.g.: $2I^- \rightarrow I_2 + 2e^-$ → Then, compare the two half-equations. If both do not have the same number of electrons, Multiply by the lowest common multiple. ◆ e.g.: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ $2I^- \rightarrow I_2 + 2e^-$ The lowest common multiple between 6 and 2 is 6, so: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ $2I^- \rightarrow I_2 + 2e^- \quad (\times 3)$ <u>Final equations:</u> $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ $6I^- \rightarrow 3I_2 + 6e^-$
C	→ Finally, Combine both half-equations to form the ionic equation: ◆ e.g.: $Cr_2O_7^{2-} + 14H^+ + \cancel{6e^-} \rightarrow 2Cr^{3+} + 7H_2O$ $+ \quad 6I^- \rightarrow 3I_2 + \cancel{6e^-}$ <u>Final equation:</u> $Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$

- Complete the ionic half-equation for the reduction of the dichromate(VI) ions to chromium(III) ions.
 - $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

❖ Disproportionation reactions:

(R.Q.): Reactions of Chlorine with NaOH

. Cold, dilute NaOH:**. Hot, Concentrated NaOH:****Reactions of Chlorine with Water****Chlorine Reaction with Water**

- Explain why a reaction is a redox reaction but not a disproportionation reaction.
 - no (element in a) single compound / molecule/species is (simultaneously) oxidised and reduced.



- Explain, in terms of oxidation numbers, what type of reaction occurs.

- disproportionation reaction, as chlorine changes from 0 in Cl_2 to +5 in NaClO_3 so oxidation, and chlorine changes from 0 in Cl_2 to -1 in NaCl so reduction.
- By considering oxidation numbers, show that this equation, $\text{Cl}_2(\text{g}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{HClO(aq)} + \text{HCl(aq)}$, represents a disproportionation reaction.
 - Cl in Cl_2 goes from 0 to (+)1 in HClO and is oxidised
 - Cl in Cl_2 goes from 0 to -1 in HCl and is reduced
- Explain why the reaction of chlorine and water is a disproportionation reaction by referring to relevant oxidation numbers.

- chlorine / Cl_2 / same species / element / atom is oxidised **and** reduced (in the same reaction).
 - chlorine is oxidised forming HOCl / OCl- / chlorate(I), changing oxidation number from 0 to +1.
 - chlorine is reduced forming HCl / Cl- / chloride, changing oxidation number from 0 to -1.
- On heating, NaClO disproportionates to form sodium chloride and sodium chlorate(V). Give the equation for this disproportionation reaction.
 - $3\text{NaClO} \rightarrow 2\text{NaCl} + \text{NaClO}_3$
- Iodate(V) ions, IO_3^- , react with iodide ions in acid solution. $\text{IO}_3^- + 5\text{I}^- + 6\text{H}^+ \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$. Explain, in terms of the oxidation numbers of iodine in the three species, why this is not a disproportionation reaction.
 - three oxidation numbers of I:
 - $\text{IO}_3^- = (+)5$
 - $\text{I}^- = -1$
 - $\text{I}_2 = 0$
 - two different species / ions / compounds (of iodine) are oxidised and reduced (to form the same species) OR there is not one species / ion / compound that is being oxidised and reduced OR 2 different oxidation states are not produced from one oxidation state (of iodine) OR only one species / oxidation state of iodine is formed.
- *Compare and contrast the reactions of chlorine with • water • cold, dilute aqueous alkali • hot, concentrated aqueous alkali. Include an equation for each reaction, stating the type of reaction and the oxidation numbers of the chlorine involved. State symbols are not required.
 - all are disproportionation reactions (because chlorine is both oxidised and reduced)
 - chlorine changes/is reduced from 0 to -1 (in Cl- in all reactions)
 - $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HOCl}$
 - $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaClO} + \text{NaCl} + \text{H}_2\text{O}$
 - $3\text{Cl}_2 + 6\text{NaOH} \rightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$
 - chlorine oxidised to +1, +1 and +5 shown in the appropriate reaction.

- Hydrogen peroxide decomposes in the presence of a catalyst. The type of reaction that occurs is a disproportionation reaction.

❖ Roman Numerals:

Roman Numerals

I = 1	VI = 6
II = 2	VII = 7
III = 3	VIII = 8
IV = 4	IX = 9
V = 5	X = 10

- State the meaning of the Roman numerals in chlorate(V) and chlorate(VII).
 - the Roman numeral is the charge the chlorine would have if the species were fully ionic / is the oxidation number of the chlorine (in the ions) OR the oxidation number of chlorine in chlorate(V) is (+)5 and in chlorate(VII) is (+)7.
 - Give the systematic name of KClO_4 .
 - potassium chlorate(VII) OR chlorate(VII) potassium.
- What is the formula of barium(II) ferrate(VI)?
 BaFeO_4
- What is the formula of potassium manganate(VI)?
 K_2MnO_4

Groups 1 and 2

❖ Trends in Group 2:

Trends down Group 2							
Reactivity	Ionic radius	Melting point	Ionization energy	Solubility		Thermal Stability	
				Hydroxides	Sulfates	Nitrates	$-\text{CO}_3^{2-}$
Increases		Decreases		Increases	Decreases	Increases	

- Several factors may affect ionisation energies: I) the number of protons increases. II) the outer electron is further from the nucleus. III) the amount of shielding increases. IV) the number of unpaired outer electrons increases. Which factors explain the **decrease** in ionisation energy as group 1 is descended? (II and III)

❖ (Trend of) Thermal Decomposition of Group 2:

(R.Q.): Explanation	the temperature of decomposition of carbonates or nitrates / thermal stability increases down the group, this is because the size of the cation increases, but has the same charge, so is less polarising (of the C-O or N-O bond)/ polarises the anion / carbonate / nitrate (ion) less.
When comparing two group 2 carbonates / nitrates	- State the trend of thermal decomposition (of either carbonates or nitrates, depends on what is asked in the question) down group 2. → If one of the nitrates/carbonates decomposes at a lower temperature: ◆ [named gp 2] ion/M^{2+} is smaller/ has smaller ionic radius OR has a higher (surface) charge density (than [other named gp 2] ion/N^{2+}) ◆ electron cloud more distorted in carbonate/nitrate ion/anion (with [named gp 2 element]) OR [named gp 2] (ion/cation) more polarising, so weak(er) carbon-oxygen/C-O/nitrogen-oxygen/N-O bond (with [named gp 2 element]). ◆ (Only mention this when asked about enthalpy change, thermal stability, or costs of the process): So less heat needed to decompose it/ the temperature required to decompose [named gp 2] nitrate/carbonate is lower. This makes the enthalpy change of [named gp 2] carbonate/nitrate less endothermic/positive, and so decreases costs of the process. → If one of the nitrates/carbonates decomposes at higher temperatures: reverse the argument.

Calcium nitrate is a white crystalline solid that decomposes when heated strongly with a Bunsen burner.



A student suggested an explanation for this decomposition.

"The calcium ion is polarised by the nitrate ion. This distorts the electron cloud around the nitrate ion, and so weakens the bond between the two ions."

Identify two errors in this suggestion and include corrections that give an explanation of the decomposition.

- *Error: calcium (ion) is polarised by the nitrate (ion)*
- *Correction: nitrate (ion) is polarised by calcium (ion)*
- *Error: weakens the bond between the two ions/between calcium and nitrate (ions)*
- *Correction: weakens the (covalent) bond(s) within the nitrate (ion).*

❖ Thermal decomposition of Group 2 vs Group 1:

Thermal Decomposition of Metal Compounds

Metal	Nitrate (NO_3)	Carbonate (CO_3)	Hydroxide (OH)
Potassium Sodium	Metal nitrate $\rightarrow$ Metal nitrite + Oxygen $2\text{NaNO}_3(\text{s}) \rightarrow 2\text{NaNO}_2(\text{s}) + \text{O}_2(\text{g})$	No decomposition	
Calcium Magnesium Aluminium Zinc Iron Lead Copper	Metal nitrate $\rightarrow$ Metal oxide + Nitrogen dioxide + Oxygen $2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$	Metal carbonate $\rightarrow$ Metal oxide + Carbon dioxide $\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$	Metal hydroxide $\rightarrow$ Metal oxide + Water $\text{Mg}(\text{OH})_2(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2\text{O}(\text{g})$
Silver Gold	Metal nitrate $\rightarrow$ Metal + Nitrogen dioxide + Oxygen $2\text{AgNO}_3(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + 2\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$	Metal carbonate $\rightarrow$ Metal + Carbon dioxide + Oxygen $\text{Ag}_2\text{CO}_3(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{CO}_2(\text{g}) + \text{O}_2(\text{g})$	Do not exist

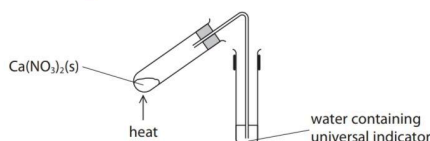
Group 1 Nitrates



Group 2 Nitrates



A sample of solid anhydrous calcium nitrate, $\text{Ca}(\text{NO}_3)_2(\text{s})$, is heated strongly for several minutes in the apparatus shown.



State two observations that would be made.

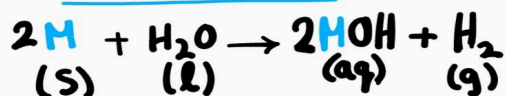
- *brown gas (produced)*
- *bubbles (through water)*
- *(water/indicator) turns red*
- *solid melts*
- *Compounds of lithium (Group 1) and compounds of magnesium (Group 2) have similar properties. Explain : ● the trend in thermal stability of the Group 2 nitrate ● why the ionic radii of Li^+ and Mg^{2+} are similar ● why the thermal decomposition of lithium nitrate is similar to that of magnesium nitrate but different from the thermal decomposition of sodium nitrate. ● describe any differences in the products of thermal decomposition of the Group 1 nitrates ● give equations for the thermal decomposition of sodium nitrate and of magnesium nitrate.
 - *thermal stability increases down Group 2*
 - *ionic radius / size of ions increases (down groups) and polarising power (of cations) decreases / charge remains the same/ $2+$, so NO breaks less easily / requires more energy to break (down groups).*
 - *magnesium has an extra shell than lithium (and as the charge on magnesium is Mg^{2+} and lithium Li^+) so the ions are similar in size.*
 - *LiNO_3 decomposes like Group 2 nitrates OR Group 1 nitrates other than lithium form (metal) nitrite/ nitrate(III)/ MnO_2*
 - *equation for the thermal decomposition of lithium or magnesium nitrate:*
 $4\text{LiNO}_3 \rightarrow 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$ OR
 $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$
 - *equation for the thermal decomposition of sodium nitrate: $\text{NaNO}_3 \rightarrow \text{NaNO}_2 + \frac{1}{2}\text{O}_2$*
 - *Lithium and magnesium nitrate produce nitrogen dioxide but sodium nitrate does not/ or sodium nitrate produces a nitrite but lithium and magnesium do not, because Mg^{2+} and/or Li^+ are smaller than Na^+ OR Mg^{2+} and/or Li^+ have greater charge density than Na^+*
 - *so polarise the nitrate ion/ anion more (causing Mg^{2+} and Li^+ to thermally decompose in a similar manner)*
 $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$
- Magnesium carbonate decomposes readily when heated but sodium carbonate does not. Explain

this observation by including reference to the charge and size of the cations.

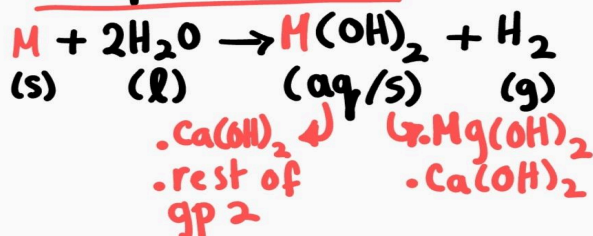
- (First mark – charge)
magnesium ion has a greater /higher charge (than sodium ion) OR magnesium is Mg^{2+} and sodium is Na^{+}
- (Second mark – size)
magnesium ion is smaller (than sodium ion) OR sodium ion is larger (than magnesium ion)
- (Third mark – comparison of polarisation)
magnesium / Mg^{2+} /cation /smaller ion causes more polarisation / distortion OR sodium / Na^{+} /cation /larger ion causes less polarisation / distortion
- (Fourth mark – what is being polarised)
 $C-O$ bonds / $C=O$ bonds.
- Suggest why a pure sample of magnesium carbonate will not be produced from the decomposition of magnesium oxalate, MgC_2O_4 , even if the sample is heated for longer.
 - the magnesium carbonate may begin to decompose (before the oxalate decomposition is complete) / the sample would be contaminated with magnesium oxide.
- State why the student used a warm oven to remove the water from the hydrated salt of magnesium nitrate, rather than direct heating with a Bunsen burner.
 - magnesium nitrate decomposes / breaks down (when heated with a Bunsen burner)
- Which cation would be expected to form the nitrate with the greatest thermal stability? Larger ionic radius and smallest charge.

❖ Reaction of Groups 1 vs Group 2:

Group 1 + Water



Group 2 + Water

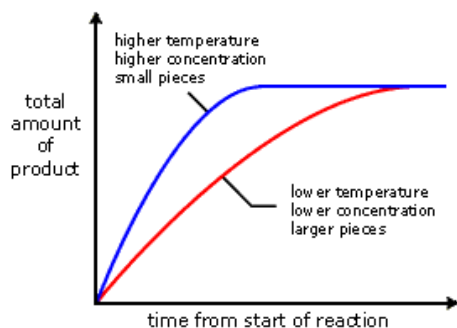


- Lithium hydroxide solution can be made by reacting lithium with water.

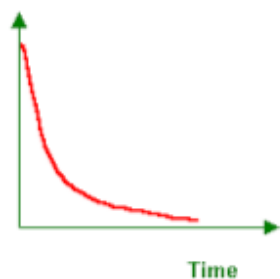
equation	observation
LITHIUM + WATER → LITHIUM HYDROXIDE + HYDROGEN $2Li + 2H_2O \rightarrow 2LiOH + H_2$	• bubbles (of gas)/ fizzing/ effervescence • Li or solid disappears/ dissolves/ disintegrates /gets smaller OR Li floats on water/moves on the water

- *Samples of calcium and magnesium are added to separate test tubes of cold water containing a few drops of universal indicator. State and explain the differences in the observations made in each experiment. Include an equation in your answer.
 - $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$
 - effervescence
 - (indicator) turns blue
 - faster / more vigorous reaction with calcium **and** linked to an observation:
 - has faster bubbling
 - changes colour faster
 - forms (white) ppt faster
 - gets hotter (Ignore more exothermic)
 - (solid) disappears faster
 - OR no (observable) reaction with magnesium
 - Ca has lower ionisation energy OR Ca has more shielding / more shells / larger atoms

- darker blue / more alkaline / higher pH with Ca and linked to greater solubility of Ca(OH)_2 [hydroxide solubility increases down group]
- Describe how drinking magnesium sulfate solution can reduce the extent of the poisoning by barium ions. Include an ionic equation with state symbols in your answer.
 - (the barium ions are removed from solution by) precipitation of insoluble barium sulfate.
 - $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
- Lithium metal is stored under oil. When a piece of lithium metal is removed from the oil, the surface quickly tarnishes. One compound formed is lithium nitride. State how lithium nitride is formed, including the formula of lithium nitride.
 - Li reacts with nitrogen from the air
 - Li_3N
- What is the main reason why sodium peroxide is formed when sodium reacts with excess oxygen? (the sodium ion has a larger ionic radius than the lithium ion)
- In the reaction between magnesium/calcium carbonate and hydrochloric acid:



Graph for volume of CO_2 produced against time



- Graph for rate of reaction against time.
- Ionic equations:

$$2\text{H}^+ + \text{CaCO}_3 \rightarrow \text{Ca}^{2+} + \text{H}_2\text{O} + \text{CO}_2$$
 Allow $2\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{H}_2\text{O} + \text{CO}_2$

❖ Flame Tests:

Flame Color	Metal Cation
Yellow	Sodium
Lilac	Potassium
Red-violet	Rubidium
Orange-red/ (brick) red / yellow-red	Calcium
Pale/Apple green	Barium
Red	Lithium and Strontium

- (R.Q.): Explain the origin of the flame colour.
- Explain how flame color forms.
 - electrons excited/promoted to a higher (energy) level/shell/orbital by absorbing (thermal) energy /heat (from (Bunsen) flame)
 - (promoted) electrons fall back (from the excited state to the ground state) /drop to a lower energy level /(sub)shell/ orbital OR electrons return to ground state
 - emitting/releasing light /photons (in the visible region) / light waves/energy / radiation (of a specific wavelength / frequency).
- Group 2 compounds give different colours in a flame test. The best explanation is that the cations have different gaps between electronic energy levels.

Group 7

❖ Trends in Group 7:

Trends down group 7					
Reactivity	Boiling temperature	electronegativity	carbon-halogen bond enthalpy	Atomic radius	electrostatic attraction between the nucleus and outer shell of electrons
(in general) decreases					
As oxidising	As reducing				

agent	agent					
decreases	increases	increases	decreases	increases	decreases	

→ The boiling temperatures of the halogens increase down Group 7 because on descending the group the (number of electrons in the molecule increase).

- (R.Q.): Explain the trend in electronegativity descending the group from Fluorine to Iodine.
 - *elements become less electronegative (down the group)*
 - *atomic radius increases OR distance between the nucleus and outer electrons increases OR there are more shells / energy levels of inner electrons between the nucleus and the outer shell electrons*
 - *the outer electron(s)/shell is further away from the (positive) nucleus, despite the increase in nuclear charge, so there is less attraction (by the nucleus with a higher charge) for the bonding electrons / shared pair of electrons / so the bonding electrons are less strongly attracted.*
- Explain why bromine has a lower electronegativity than chlorine but a higher electronegativity than selenium.
 - *(bromine has)*
 - *one more shell (of electrons) than chlorine OR more shielding/repulsion by inner electrons than chlorine OR a larger atomic radius than chlorine, so lower attraction for (bonded/shared/a pair of) electron(s) than chlorine.*
 - *one more proton (in the nucleus) than selenium (and the same/similar shielding), so higher attraction for (bonded/shared/a pair of) electron(s) than selenium.*
- State the meaning of the term electronegativity.
 - *the ability of an atom to attract a bonding/shared pair of electrons (in a covalent bond).*

❖ Reactions of Halide ions with Silver nitrate:

Ion	Silver Nitrate	Dilute Ammonia	Concentrated Ammonia
F ⁻	No visible change (clear solution)		
Cl ⁻	White Precipitate	Colourless Solution	
Br ⁻	Cream Precipitate	Cream Precipitate	Colourless Solution
I ⁻	Yellow Precipitate	Yellow Precipitate	Yellow Precipitate

→ Ionic eqns with silver nitrate:



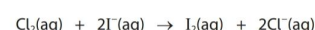
- Describe a chemical test on a solution of these crystals to confirm that the impurities present are chloride ions rather than bromide ions.
 - *add (dilute) nitric acid and silver nitrate (solution)*
 - *white precipitate forms / precipitate forms whose colour is difficult to distinguish (between white and cream)*
 - *which dissolves in dilute ammonia / dilute NH₃ / NH₃(aq)*

❖ Displacement Reactions of Halogens:

All Aqueous	Chlorine (green)	Bromine (yellow)	Iodine (Brown)
Chloride		No visible change	No visible change
Bromide	Yellow solution $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$		No visible change
Iodide	Brown solution or black precipitate $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$	Brown solution or black precipitate $\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$	

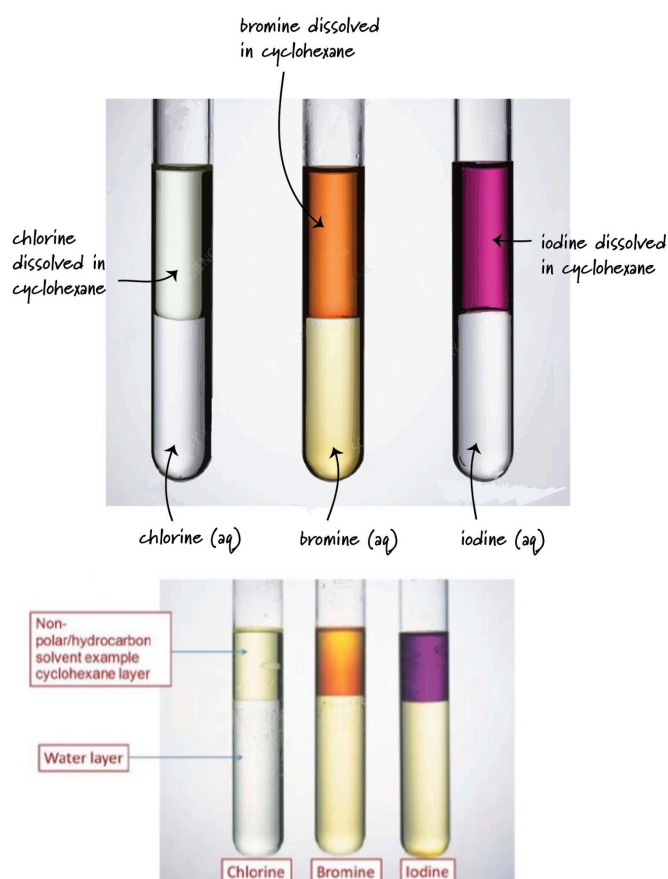
→ Which equation shows a redox reaction that would **not** be expected to occur, based on the trend in reactivity of the halogens? $2\text{F}^-(\text{aq}) + \text{At}_2(\text{aq}) \rightarrow 2\text{At}^-(\text{aq}) + \text{F}_2(\text{aq})$.

The equation for the reaction between chlorine and iodide ions is shown.



→ (R.Q.): Iodide ions reduce chlorine, and iodide ions are oxidized.

❖ Dissolving Halogens in Cyclohexane (non-polar solvents):



❖ Reactions of Sodium/Potassium halide with Conc. H_2SO_4 :

Halide	Products	Observation	Reaction	Equation
Chloride	HCl	Steamy Fumes	Acid/Base	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{HCl} + \text{NaHSO}_4$
	HBr	Steamy Fumes	Acid/Base	$\text{H}_2\text{SO}_4 + \text{NaBr} \rightarrow \text{HBr} + \text{NaHSO}_4$
Bromide	Br_2	Brown Fumes	Oxidation of Br^-	$\text{H}_2\text{SO}_4 + 4\text{H}^+ + 2\text{Br}^- \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
	SO_2	Colourless Gas	Reduction of H_2SO_4	
Iodide	HI	Steamy Fumes	Acid/Base	$\text{H}_2\text{SO}_4 + \text{NaI} \rightarrow \text{HI} + \text{NaHSO}_4$
	I_2	Purple Fumes	Oxidation of I^-	$\text{H}_2\text{SO}_4 + 4\text{H}^+ + 2\text{I}^- \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
	SO_2	Colourless Gas	Reduction of H_2SO_4	
	S	Yellow Solid	Oxidation of I^-	$\text{H}_2\text{SO}_4 + 6\text{H}^+ + 6\text{I}^- \rightarrow 3\text{I}_2 + \text{S} + 4\text{H}_2\text{O}$
	H_2S	Rotten Egg Smell	Reduction of H_2SO_4	

- Explain the difference in the reaction of concentrated sulfuric acid with chloride ions and with iodide ions.
 - iodide (ions) are better reducing agents (than chloride ions), because electrons are more easily lost from iodide (ions than from chloride ions).
- Concentrated sulfuric acid is added to solid potassium bromide. State **two** observations.
 - misty/steamy fumes (of HBr)
 - brown (fumes/gas of Br_2)
 - choking fumes/gas/smell (of SO_2)

- solid/KBr disappears/dissolves.
- (R.Q.): *Concentrated sulfuric acid reacts with solid potassium chloride and with solid potassium iodide. Compare and contrast these reactions to show the difference in reducing ability of these two halide ions. In your answer describe what would be observed and identify the products formed.
 - Potassium chloride
 - (misty/steamy fumes of) HCl/hydrogen chloride
 - Potassium iodide
 - (misty/steamy fumes of) HI /hydrogen iodide
 - purple vapour/black solid means formation of I_2 /iodine
 - bad/rotten eggs smell means formation of H_2S /hydrogen sulfide OR yellow solid means formation of S/sulfur OR Choking gas means formation of SO_2 /sulfur dioxide
 - KCl is not a redox reaction and KI is a redox reaction / cannot reduce sulfuric acid but KI can since oxidation number of chlorine remains unchanged (in the reaction with sulfuric acid) OR oxidation number of sulfur remains unchanged (in the reaction with chloride), so iodide is stronger reducing agent than chloride.
 - (because) I^- changes from -1 to 0 (in I_2), S in H_2SO_4 is reduced from $+6$ to -2 in H_2S OR S is reduced from $+6$ to 0 in S OR S is reduced from $+6$ to $+4$ in SO_2 OR Any balanced equation making H_2S , SO_2 , or S showing electrons eg :
 - $8\text{H}^+ + \text{H}_2\text{SO}_4 + 8\text{e}^- \rightarrow \text{H}_2\text{S} + 4\text{H}_2\text{O}$
 - $2\text{H}^+ + \text{H}_2\text{SO}_4 + 2\text{e}^- \rightarrow \text{SO}_2 + 2\text{H}_2\text{O}$
 - $6\text{H}^+ + \text{H}_2\text{SO}_4 + 6\text{e}^- \rightarrow \text{S} + 4\text{H}_2\text{O}$
 - $8\text{I}^- + 8\text{H}^+ + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$
- (R.Q.): Compare and contrast the reactions of concentrated sulfuric acid with solid sodium/potassium chloride and with solid sodium/potassium bromide.
 - (Similarities)
 - both produce hydrogen halides [$\text{NaX} + \text{H}_2\text{SO}_4 \rightarrow \text{HX} + \text{NaHSO}_4$ / $2\text{NaX} + \text{H}_2\text{SO}_4 \rightarrow 2\text{HX} + \text{Na}_2\text{SO}_4$]

and give off misty/steamy/white fumes.

- formation of hydrogen halides is protonation

○ (Differences)

- only the reaction of sodium bromide is redox / bromide ions are stronger reducing agents than chloride ions.
- (red-) brown fumes / gas / liquid produced with sodium/potassium bromide.
- sulfur dioxide (colourless choking gas) / bromine given off with sodium/potassium bromide $[2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}]$

→ What are the gaseous products formed, other than water vapour, when concentrated sulfuric acid is added to potassium bromide? (bromine, hydrogen bromide and sulfur dioxide only, but does not form H_2S)

→ (R.Q.): Concentrated sulfuric acid is an oxidizing agent when it reacts with sodium iodide.

Hydrogen halide	Compound produced with the lowest oxidation number of sulfur
(HBr)	sulfur dioxide / SO_2
(HI)	hydrogen sulfide / H_2S

Titration and Qualitative Analysis

❖ Titration:

CHEMISTRY TECHNIQUES: TITRATION
Used to determine the concentration of a particular solution, by measuring how much of a solution of known concentration reacts with a known volume of it.

EQUIPMENT
BURETTE
PIPETTE
CONICAL FLASK
PIPETTE FILLER
INDICATOR

USING A PIPETTE
Pipettes should never be filled direct from stock bottles of solution as this could contaminate them. Instead, fill them from some of the solution poured into a beaker. They're calibrated to allow for a small amount of solution remaining in the tip after the correct volume has been delivered, so this remainder shouldn't be forced out.

FILLING THE BURETTE
As with the pipette, the burette should be rinsed with distilled water then the solution it is to be filled with before filling. It is necessary that the area under the stop also contains solution as the burette's scale includes this volume. Without doing this, the volume recorded for titrations would be higher than the actual value.

USING INDICATORS
In acid-base titrations, a range of indicators can be used. These are solutions which change colour at a specific pH, and can be used to precisely identify when the neutralisation reaction is complete (the end point). Different indicators are suitable for different acid-base combinations.

CARRYING OUT THE TITRATION
To carry out the titration, the tap of the burette is opened to allow the solution inside to flow into a known volume of the solution in the conical flask. The amount of solution from the burette required to reach the end point is recorded. A rough titration is usually followed by more accurate runs. Multiple titrations are carried out until concordant titres are obtained (within 0.20 cm³ of each other).

CARRYING OUT CALCULATIONS
 $n = cv$
NO. OF MOLES = CONCENTRATION × VOLUME
1. Calculate number of moles of solution added from the burette.
2. Determine the number of moles of solution in the conical flask using the equation for the reaction and reacting ratios.
3. Calculate the concentration of the solution in the conical flask by rearranging the equation $n = cv$.

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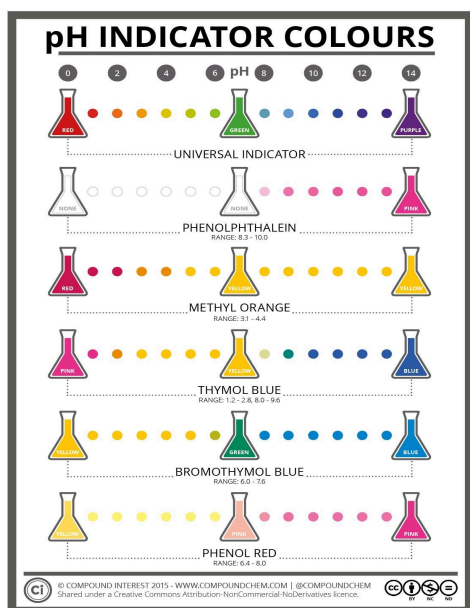
Detailed Steps

1. rinse glassware, such as a conical flask with appropriate solutions [distilled water then acid/alkali (depending on what will be added in the beaker)].
2. Fill the burette with the alkali solution, ensuring there are no air bubbles.
3. use a pipette and pipette filler to transfer 25 cm³ / 10 cm³ of acid to a conical flask
4. add indicator to the acid in the conical flask and carry out a range finder/rough titration
5. add alkali drop by drop near the end point
6. repeat titrations until concordant/within ± 0.2 cm³

Rough outline of steps

- titrate a known volume of alkali, with a (known concentration of) acid, until a colour change is seen

Indicators



INDICATOR	COLOUR IN ACID	COLOUR IN ALKALI	ACID-BASE COMBINATION
methyl orange	red	yellow	strong acid - weak base and strong acid - strong base
phenolphthalein	colourless	pink	weak acid - strong base and strong acid - strong base

Errors in titration

- first titre greater than the rest:
- ◆ First titre likely to be a rangefinder / rough titration / estimate (so done quickly).
 - ◆ There was an air bubble (in the burette jet which fills before the titration starts).
 - ◆ Burette rinsed with water (rather than alkali (or acid, depending on what is in the burette)).

Using metal hydroxides

- When a metal hydroxide is more soluble than another, the concentration of hydroxide ions will be greater, so the (mean) titre value would be bigger.

- Another student suggested that the reaction between manganate(VII) and rhubarb juice could be used to find the concentration of ethanedioic acid in rhubarb juice. Outline an experiment that could be carried out to find this concentration.
 - titrate a known volume of (undiluted) rhubarb juice
 - with a known concentration of manganate(VII)

- until a colour change is seen

Another student carried out the same experiment but obtained a different answer. The method this student used for preparing the sodium carbonate solution is shown.

I weighed 1.00g of hydrated sodium carbonate in a weighing bottle and then tipped the solid into a 250 cm³ volumetric flask.

I dissolved the solid in a small amount of deionised water and then added deionised water up to the mark.

I then carried out a series of titrations.

- Identify two errors that the student made in preparing this solution. Include the effect these errors will have on the titration volumes, with a reason.
 - (First error) not washing the weighing bottle (with deionised water) OR not re-weighing the weighing bottle
 - (First Effect with reason) the titration volume is less, because lower/decreased concentration / more dilute
 - (Second error) not shaking/inverting/mixing the solution in the volumetric flask
 - (Second effect with reason) titres inconsistent/varied, because non-homogeneous solution OR titre volumes may be smaller in parts of the solution where the concentration is lower.
- Give a reason why a further titration was not carried out.
 - two concordant titres already obtained



Fig 1. Neutralisation.

❖ Weighing by Difference (to find Atomic Mass):

Steps

- For example, you want to identify the element in metal carbonate, MCO_3 . The steps:
1. weigh the test tube (empty) and (reweigh) after adding some (solid) carbonate.
 2. heat (the solid in the test tube using the Bunsen burner) until there is no change in mass.

3. calculate moles of carbon dioxide (which is equal to the moles of carbonate/metal oxide) from the loss in mass and molar mass of carbon dioxide.
4. calculate the molar mass of carbonate from (initial) mass OR OR calculate the molar mass/Mr of metal oxide from (remaining) mass and moles (of carbonate/metal oxide/ carbon dioxide).
5. calculate the molar mass/Ar of element (allowing its identification) from subtraction of 60/CO₃ from the molar mass/Mr of metal carbonate) / from subtraction of 16/O from the molar mass/Mr of metal oxide.

Errors

- if the metal is coated with something, such as an oil, and it is not removed beforehand, this coating will be weighed with the mass of the metal, and so the relative atomic mass of the metal will be greater than its real/actual Ar.

- State how you could ensure that all the water had been removed from the crystals.

- **heat to constant mass /until no change in mass.**

- A student investigated the role of the catalyst in this reaction.

Procedure

Step 1 Heat a sample of KClO₃ , in a test tube, with a known mass of insoluble catalyst until the mass remains constant.

Step 2 Mix the contents of the test tube with water.

Step 3 Filter the mixture and rinse with deionised water.

Step 4 Dry the remaining solid.

Step 5 Measure the mass of the dry solid.

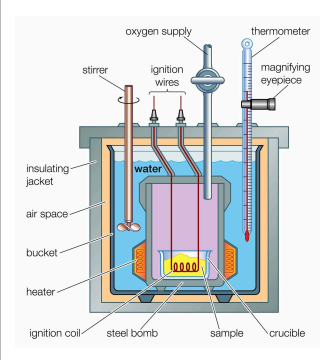
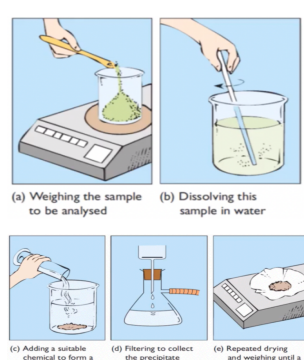
- Explain how each of the steps in this procedure is needed to show that the catalyst is not used up in this reaction.

- **heat to constant mass so all of the potassium chlorate(V) decomposes.**
 - **the solid product or potassium chloride dissolves (when the water is added).**
 - **the rinsing removes potassium chloride (solution, which would otherwise add to the mass of the solid when it dries.**
 - **drying ensures the final mass recorded is only that of that catalyst.**

- **the mass (of solid) recorded (at the end of the procedure) should be the same of that of the catalyst at the start.**

- Pair of solution that would give greatest mass of precipitate = **pairs of solutions that will form two insoluble precipitation** (e.g. BaCl₂ and Ag₂SO₄ ,when mixed together, will form BaSO₄ and AgCl)

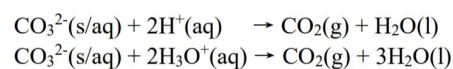
❖ **Different methods to determine concentrations of a solution, other than using titration:**

Calorimetric method	Gravimetric method
	
→ Errors: value obtained may be lower than expected since heat may be lost to the surroundings /absorbed by the container or apparatus	→ Errors: value obtained may be higher than expected since the product may have been wet/ have impurities.

❖ **Qualitative Analysis (Tests):**

Test	Result
Warm NH ₄ Cl (or ammonium compounds) with aqueous NaOH solution	an alkaline gas (ammonia), which turned damp red litmus paper blue , evolved.

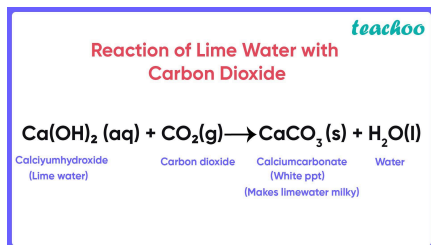
- Nitric acid used as test reagent to identify hco₃⁻ ions.
- (R.Q.): Carbonate ions may be identified by their reaction with aqueous acid to produce carbon dioxide. Give the ionic equation for this reaction. Include state symbols.



○

- (R.Q.): State what would be seen when carbon dioxide is bubbled through limewater, and write its chemical equation.

- the mixture/solution would go cloudy/milky/ a white precipitate would form.



-

Colour of nitrogen dioxide	Colour change of damp litmus paper
brown	blue to red

❖ Safety Precautions:

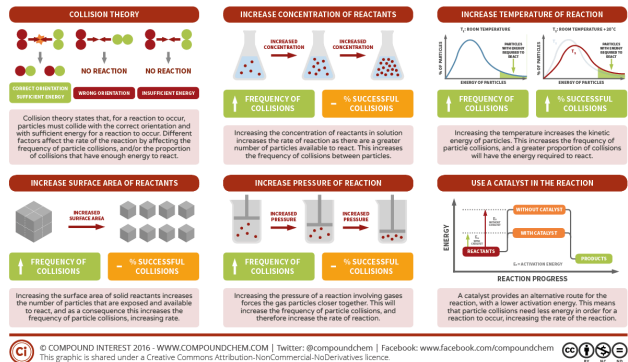
Scenario	Precaution	Reason
Dealing with gases (e.g. NO ₂ , CO, HCl, Cl ₂ , etc.) by opening a bottle, or thermal decomposition of solid produces these gases	• Use fume cupboard. • Open bottle in a well-ventilated laboratory/outside. • wear a gas mask	• these named gases are toxic
Dealing with (strong) acids	• Wear (safety) glasses/goggles/spectacles. • Wear gloves and a lab coat.	• These acids are corrosive

Kinetics

❖ Rate of Reactions:

MAKING CHEMICAL REACTIONS HAPPEN FASTER

There are a number of different things that we can change to make a chemical reaction faster. Here, we explain the concept of collision theory, and how it can be used to explain the effects of five different factors on the rate of a chemical reaction.



(R.Q.): Effect of Catalyst

- a catalyst decreases/reduces the activation energy required by introducing/providing an alternative pathway/route/mechanism for energy, so a larger/greater proportion of / larger number of / more molecules/particles have energy $\geq E_a$ / energy required for the reaction to take place/area under the curve to the right of /at a greater energy than E_a in a Maxwell-Boltzmann curve increases/is larger, and the frequency/number per unit time of successful collisions increases/higher proportion of/more collisions are successful, thus the rate of reaction increases.

Effect of concentration

- When increasing the concentration: more particles will be in the same volume, thus increasing the number of (successful) collisions per unit time.

Effect of temperature

- When increasing the temperature: the molecules/particles have more kinetic energy, so more molecules/particles have energy $\geq E_a$ and more successful collisions occur per unit time, thus the rate of reaction increases.

- Magnesium metal reacts with hydrochloric acid. Which change in condition would have no effect

on the initial rate of this reaction? (an increase in the volume of acid solution).

- Increasing pressure and/or temperature increases rate.
- the units for the rate of reaction is $\text{mol dm}^{-3} \text{s}^{-1}$
- Which statement best explains why the rate of a chemical reaction increases with increasing temperature? the frequency of collisions with $E \geq E_a$ increases.

❖ Maxwell-Boltzmann Energy Distribution Curves:

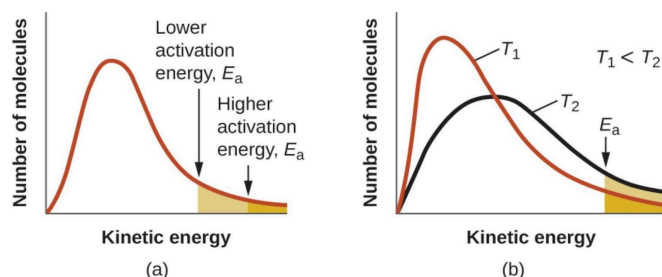


Figure 4. (a) As the activation energy of a reaction decreases, the number of molecules with at least this much energy increases, as shown by the shaded areas. (b) At a higher temperature, T_2 , more molecules have kinetic energies greater than E_a , as shown by the yellow shaded area.

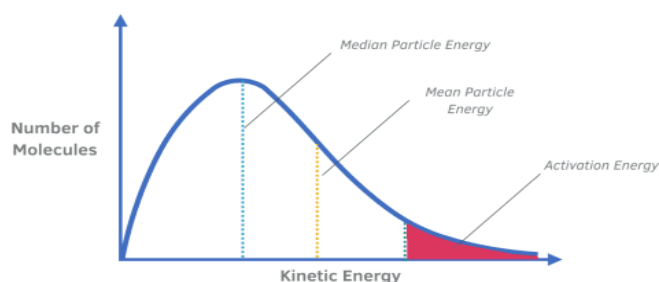
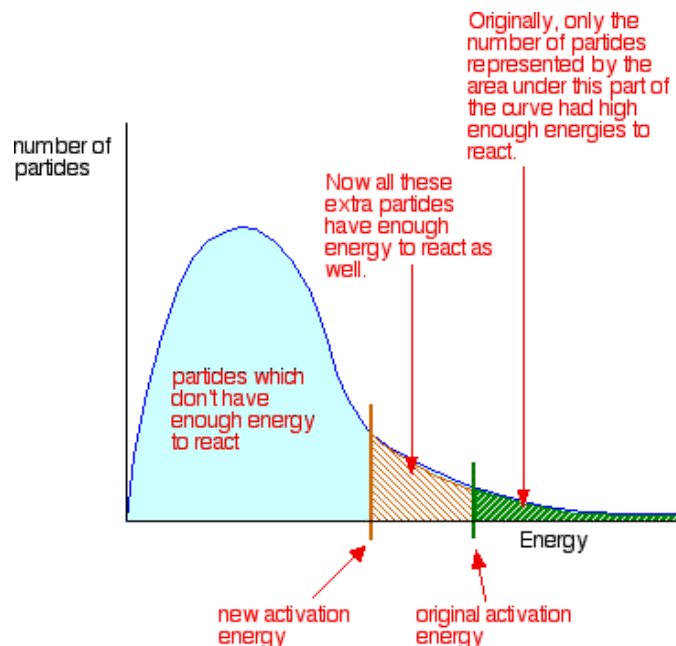
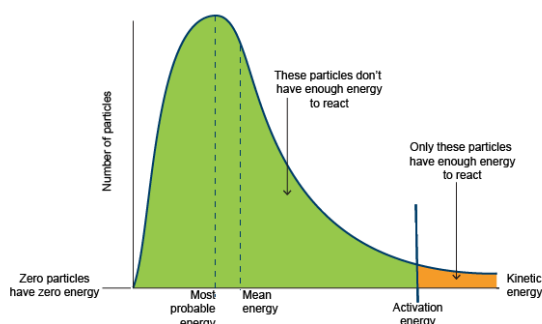
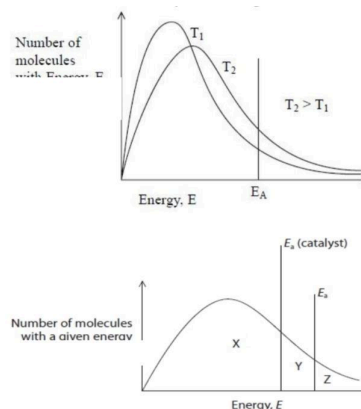


Fig 1. Maxwell-Boltzmann Distribution. This figure shows a Maxwell-Boltzmann Distribution and you should be able to draw, label and interpret one yourself. Additionally, you should be able to describe what is happening at each stage of the plot.



- *Process B involves gas molecules reacting at high temperature in the presence of a catalyst. Explain the effects on the rate of reaction of increasing the temperature and using a catalyst in Process B, referring to a labelled diagram of the Maxwell-Boltzmann distribution.



O

- molecules with $E \geq E_a$ can react/ collisions are successful
- increasing temperature (increases energy of all molecules so) increases molecules / collisions with $E \geq E_a$ (so rate increases)
- adding a catalyst (provides an alternative pathway which) lowers E_a so more molecules / collisions have $E \geq E_a$

- (R.Q.): Using the Maxwell-Boltzmann distributions, explain why increasing the temperature and adding a catalyst both increase the rate of reaction.

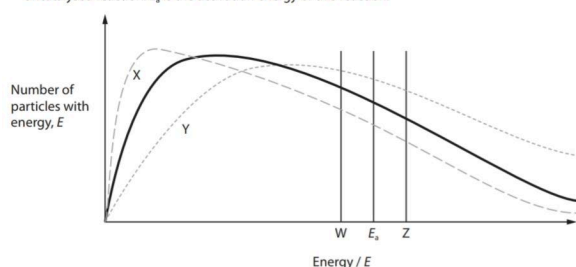
○ EITHER

- Increasing the temperature increases the (average kinetic) energy of collisions **and** so more collisions have energy greater than the activation energy
- adding a catalyst lowers the activation energy **and** so more collisions have energy greater than the activation energy

○ OR

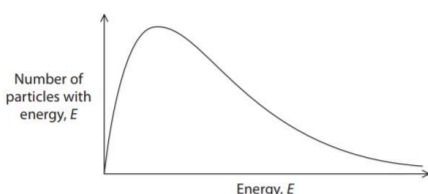
- (rate increases when) more collisions have an energy greater than (or equal to) the activation energy
- the area under the curve beyond E_a is increased when the temperature is increased **and** when E_a becomes E_{cat} .

4 The solid line on the graph below shows the Maxwell-Boltzmann distribution for an uncatalysed reaction. E_a is the activation energy of this reaction.



Which row shows the correct Maxwell-Boltzmann curve and activation energy for the reaction at a higher temperature with a catalyst? Y and W.

The distribution of molecular energies for a sample of gas in a sealed container is shown.



Why does the distribution of energies start at the origin, (0,0)? all molecules possess some energy.

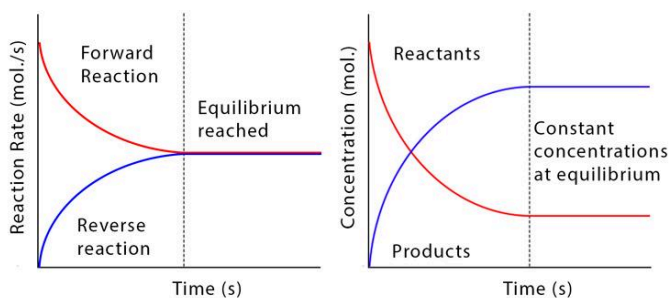
Some of the gas is removed and then the container is resealed and the gas is cooled. How does the new distribution of molecular energies compare to the original sample? (Area under curve decreases, position of peak shifts to the left).

❖ Temperature, Pressure, Catalysts, and Sustainability:

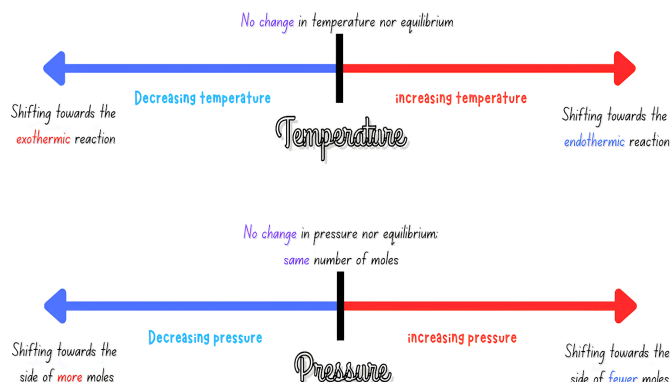
- An alternative method for producing methanal uses a silver catalyst at a temperature of 650°C. The reaction is shown. $\text{CH}_3\text{OH(g)} \rightarrow \text{HCHO(g)} + \text{H}_2\text{(g)}$. Suggest one advantage and one disadvantage of this method, other than the cost of the silver catalyst.
 - Advantage: hydrogen is a useful by-product
 - Disadvantage: higher temperature means higher energy costs.
- Give one other reason for not operating the process at a pressure greater than 7MPa. Do not include safety considerations in your answer.
 - higher pressures require special plant / equipment to withstand the higher pressures (which are expensive)
- Many different catalysts have been used in a reaction, including MoO_3 , WO_3 and Re. State two characteristics that all these catalysts have in common.
 - (all contain) d-block elements/(all contain) transition metals
 - (all are) heterogeneous (catalysts)/ in a different phase/state to reactants, and are solids.
- (R.Q.): Explain how the use of a catalyst makes this reaction more sustainable.
- Give a reason why using a catalyst to increase the rate can make a process more sustainable.
 - (a catalyst) (operates/works at) a lower temperature/ allows a lower temperature to be used, less energy required/needed (for the reaction to proceed at an acceptable rate) OR less burning of fossil fuels / less greenhouse gas evolved / less CO_2 evolved, and is not used up/can be used again.
 - (a (specific) catalyst is specific for molecule) so there is less waste/fewer other products/improves atom economy.

(Dynamic) Equilibrium❖ **Equilibrium:****Characteristics of dynamic equilibrium**

- the concentration of reactants and products is constant
- the forward and backward/reverse reactions occur at equal rates
- the equilibrium can be reached from either direction
- the position of equilibrium is affected by temperature, by pressure and by concentration.
- the total number of molecules is constant.

Dynamic Equilibrium

ChemistryLearner.com

Equilibrium**Equilibrium and Le Chatelier's principle**

Reversible chemical reactions reach equilibrium in closed systems (no substances added or lost). Here's how different conditions affect that equilibrium.

Equilibrium	Concentration	Temperature	Pressure
Equilibrium $A + B \rightleftharpoons C + D$	Reactant concentration increased The equilibrium position shifts to reduce the reactant concentration. # ↓	Temperature increased ↑ °C The equilibrium position shifts to reduce the temperature. ↓ °C	Pressure increased ↑ Pa The equilibrium position shifts to reduce the pressure. ↓ Pa
In reversible reactions products of the reaction can react to produce the original reactants. At dynamic equilibrium the rates of the forward and backward reactions are equal, the concentrations of the reactants and products don't change. If you remove all of the same substances from a system, the size of the hole and pile of dirt stay the same.	Reaction forming products favoured In the example above, the new equilibrium mixture will contain more A and B, and less C and D. # ↑	The endothermic reaction will be favoured In the example above, the new equilibrium mixture will contain more A and B, and less C and D. # ↑	Side of reaction with fewer gas molecules favoured In the example above, the new equilibrium mixture will contain more C and D, and less A and B.
Le Chatelier's principle An analogy for changing equilibrium conditions. Le Chatelier's principle states that when a change is made to the conditions of a dynamic equilibrium, the system moves to counteract the change, causing changes in quantities of reactants and products.	Product concentration increased The equilibrium position shifts to reduce the product concentration. # ↓	Temperature decreased ↓ °C The equilibrium position shifts to increase the temperature. ↑ °C	Pressure decreased ↓ Pa The equilibrium position shifts to increase the pressure. ↑ Pa
Reaction forming reactants favoured In the example above, the new equilibrium mixture will contain more C and D, and less A and B. # ↓	The exothermic reaction will be favoured In the example above, the new equilibrium mixture will contain more C and D, and less A and B. # ↓	Side of reaction with more gas molecules favoured In the example above, the new equilibrium mixture will contain more A and B, and less C and D.	

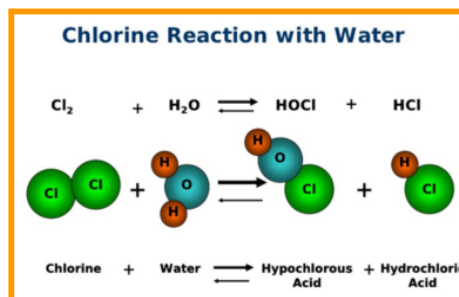
Note: using a catalyst increases the rate of both the forwards and backwards reactions but doesn't change the equilibrium position.

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- There is no change in an equilibrium if a catalyst is added to the reaction mixture.



- State and explain what happens to the position of equilibrium when sodium hydroxide solution, NaOH(aq), is added to this equilibrium mixture.
 - shifting eqm to the right (to make more HCl/HOCl)
 - because (NaOH would) react/ remove the HCl/H⁺ (and HOCl).
- Explain why a manufacturer might suggest a pressure of greater than the compromise pressure of 7MPa in hydration of ethene.
 - the equilibrium shifts giving a greater yield of ethanol
 - (because) there are fewer moles (of gas) on the products side / right hand side (than there are on the reactants side / left hand side).

Ammonium nitrate, NH_4NO_3 , is used in the manufacture of fertilisers and explosives. It is produced on a large scale using only methane, water and air. The process has four stages.

(a) The first two reactions in Stage 1 involve the production of hydrogen.

At temperature T_1 , methane reacts with excess steam to give hydrogen.



At a different temperature, T_2 , the carbon monoxide reacts with more steam.



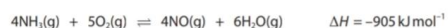
Give the reason why excess steam is used in the first reaction.

- shifts position of equilibrium to the right
OR increases the (equilibrium) yield (of H₂)

Predict which of T1 and T2 is the higher temperature. Justify your answer.

- T1 (is higher) and (first reaction is) endothermic.

(e) In Stage 3, nitrogen monoxide, NO, is produced in the reaction between NH₃ (from Stage 2) and O₂ (from the air). The conditions used are a temperature of 1100 K in the presence of a platinum-rhodium catalyst.



Give one reason why a high temperature is needed in this reaction.

- increase rate OR rate is slow at low temperature OR catalyst does not work at low temperature OR so more reactants/collisions have E equal to or greater than E_a OR to break O=O/NH bonds.

Suggest why only a small amount of energy is used to maintain the temperature at 1100 K.

- (forward reaction is highly) exothermic OR (forward reaction) releases (a lot of) heat (energy)

The NO from the first reaction in Stage 3 is cooled and then converted to nitrogen dioxide, NO₂, by reaction with more O₂.



Nitric acid, HNO₃(aq), is produced by the addition of water.



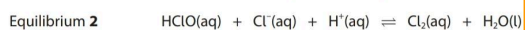
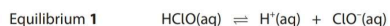
Explain how adding water in the second reaction affects the yield of NO₂ in the first reaction.

- NO₂ removed (in second reaction)
- shifting position of equilibrium (in first reaction) to right and increasing the yield (of NO₂)

(b) Bleach, NaClO(aq), is formed by the reaction of chlorine with cold, dilute aqueous sodium hydroxide.



The following species and equilibria are present in solutions of NaClO(aq).



Household bleach is stored at a pH between 11 and 13.

Household bleach should not be stored near strong acid.

Explain why mixing strong acid with bleach is dangerous. Justify your answer using the equilibria shown.

- equilibrium 1 shifts left (to make HClO)

- equilibrium 2 shifts to the right (to make chlorine solution)
- chlorine gas (produced) (which is) toxic.



Toilet cleaners that remove limescale should not be mixed with chlorine-based disinfectants.

Explain, using the relevant equation, why these two types of toilet cleaner should not be mixed.

- toxic chlorine would be produced
- (because) the chlorine disinfectant equilibrium would move to the left (because the concentration of H⁺ would increase).

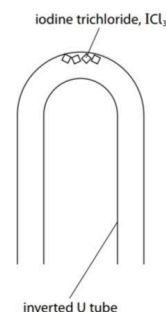


The conditions for this industrial reaction are a pressure of 50 atm, a temperature of 250 °C and a copper catalyst.

Explain why high pressure is used.

- EQUILIBRIUM
 - there are more (gaseous) moles/molecules on LHS / less (gaseous) moles on RHS
 - (so high pressure) moves equilibrium to RHS / increases yield of methanol
- OR KINETIC
 - an increase in pressure increases the number/rate of collisions (of molecules)
 - (So) increases the rate of reaction,

(ii) Air has an average density of 1.25 g dm⁻³. If the U-tube used in 23(d) is inverted, as shown in the diagram, the solid yellow iodine trichloride produced in the equilibrium reaction turns to a brown liquid.



Explain this observation.

- chlorine (gas) is more dense than air
- chlorine (gas) removed (from the equilibrium)

- *position of equilibrium moves to the LHS (more brown liquid/ICl).*

Phosgene can be formed from carbon monoxide and chlorine, using a catalyst of activated carbon.



State and explain how the reaction conditions could be changed to maximise the **equilibrium** yield of phosgene in this reaction.

- *decrease the temperature, as the (forward) reaction is exothermic.*
- *increase the pressure, as there are fewer moles of (gas) on the product side.*
- *EITHER: remove the phosgene (as it is formed) to reduce the concentration of product (so equilibrium moves to the right) OR: add more CO / Cl₂ to increase the concentration of the reactants (so equilibrium moves to the right)*

❖ Compromise Temperature:

- The forward reaction is exothermic in a reaction, and a catalyst is also used. Lower temperatures increase yield of forward reaction, but catalysts may not work at lower temperature + rate of reaction will be slower, so a compromise temperature is used between rate and yield, to ensure the rate is fast enough and to be economically viable.
- If only a small amount of energy is used to maintain the high temperature of a reaction, then this reaction is highly exothermic.

Methanol can be produced from the captured carbon dioxide by reaction with hydrogen using a copper catalyst. The equation for the reaction is shown.



Explain why the conditions for the industrial production of methanol are a pressure of 50 atmospheres and a temperature of 250°C.

- *the (forward) reaction is exothermic so a low(er) temperature increases the yield of methanol, but the rate will be too low (so a compromise temperature is used).*
- *there are fewer moles of gas on the RHS/products, so high pressure will move the equilibrium position to the RHS/increase the yield of methanol.*

- Explain why the temperature used in the hydration of ethene is a compromise.

- *the forward reaction is exothermic so is favoured by a lower temperature (giving a higher yield)*
- *but the lower temperature results in a rate of reaction that is too slow (to be economically viable).*

Ethanol can be made in industry by the reaction of ethene with steam, using a phosphoric(V) acid catalyst.



The reaction is carried out at 300°C and 60 atm. An initial yield of 5 % is achieved when the ethene and steam first pass through the reactor.

- *Explain the chemical reasons for the conditions used and why such a low initial yield is acceptable in the industrial process.
 - *high temperature increases rate as more particles have $E \geq E_a$*
 - *Catalyst increases rate by providing alternative mechanism / catalysts lower activation energy*
 - *but high temperature reduces yield / moves eqm to LHS as reaction is exothermic*
 - *so (high) temperature (of 300°C) is a compromise (between rate and yield)*
 - *high pressure increases the yield as reaction / equilibrium moves to side of fewest particles / high pressure increases rate as more particles in the same volume*
 - *(low yield acceptable) as unconverted reactants can be recycled / passed through reactor again.*
- Discuss the factors leading to a choice of 250°C in the reaction between carbon dioxide and hydrogen to form methanol.
 - *the (forward) reaction is exothermic*
 - *a lower temperature would increase the yield / move equilibrium to RHS or reverse argument*
 - *a (compromise) temperature is used to ensure the rate is fast enough (without the equilibrium position moving too far to the LHS)*
ALLOW (increased/high/stated) temperature increases the rate of reaction.

❖ Advantages and Disadvantages of using High Temperature and Pressure:

Advantages	- increased rate of reaction - increased temperature leads to more frequent successful collisions / particles increase in kinetic energy, as well as more particles have $E \geq E_a$, and increase in pressure means more particles per unit volume / more frequent successful collisions / increases occupation of catalyst active sites (if a catalyst is used). - increased pressure increases yield, as equilibrium moves to the RHS/product side, as this has the fewer gas molecules / particles / moles. (If product side has fewer moles).
Disadvantages	- increased temperature reduces yield as reaction is exothermic and equilibrium moves to left hand side/reactant side, and increases energy costs. - high pressure increases costs of equipment since it requires expensive/specialist equipment to withstand high pressure, and requires more energy, so increases cost. - high temperature and high pressure used as a compromise between advantages and disadvantages.

- Increasing the amount of water in the reaction mixture, without changing the pressure, might also be suggested. Suggest one advantage and one disadvantage of increasing the amount of water.
 - (an advantage is that) it pushes equilibrium to the right increasing the yield (of ethanol)
 - (a disadvantage is) that there is more water condensed with the ethanol (to separate) OR the catalyst is washed from the support / is diluted by the water / dissolves in the condensed water and is removed from the reaction vessel.

A one-step process is being developed. This is carried out at 101 kPa and 800°C.



- Deduce how the reaction conditions could be changed to increase the yield, explaining why such changes may not be economical.
 - the reaction is exothermic, so (by Le Chatelier) the temperature could be

decreased (to increase the yield), but this would decrease the rate.

- pressure could be increased as fewer moles of gas on the right-hand side, but (increasing the pressure would be more expensive because either) increasing the pressure uses more energy OR equipment walls need to be thicker/stronger.

(e) Ethanol can also be produced by the hydration of ethene.



*(i) Typical conditions are 300°C and 60 atm with a catalyst of phosphoric acid.

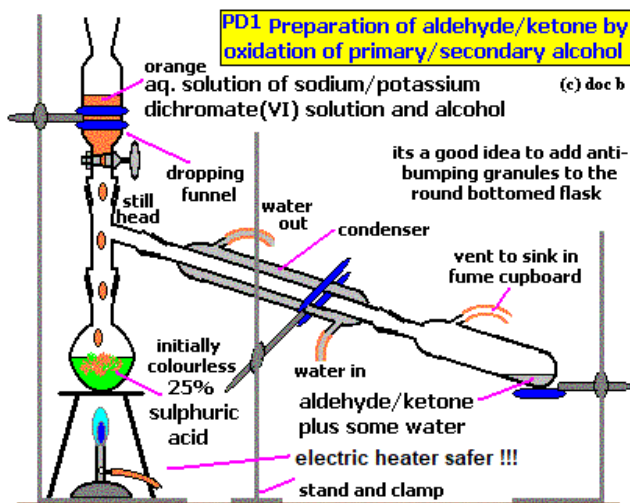
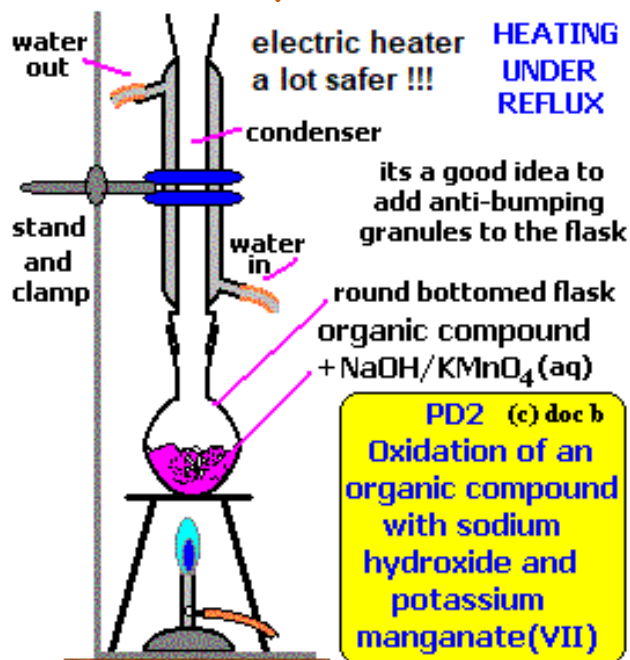
- *Explain why these conditions are used, by describing the effect of changing the temperature and pressure on rate of reaction, equilibrium yield and cost.
 - if temperature increases, rate increases (because more particles have energy greater than E_a)
 - if temperature increases, yield decreases (because forward reaction is exothermic/ reaction moves in the endothermic direction)
 - if pressure increases, rate increases (because particles are more crowded and) collision frequency increases)
 - if pressure increases, yield increases (as there are fewer particles on the RHS of the equation)
 - the catalyst allows the use of lower temperature/less energy (for the same rate)
 - high pressure (and temperature) are very expensive.

(R.Q.): Under these conditions, only about 5% of the ethene is converted into ethanol as it passes over the catalyst, at equilibrium, but the overall yield of ethanol for the process is 95%. Suggest how the overall yield of this process can be improved to make it economically viable.

- the reaction is cooled so that ethanol (and excess water) condenses / liquefies OR the ethanol is removed from the reaction mixture.
- the unused ethene / starting materials / reactants (which are still gaseous) are recycled.

Intro to Organic Chemistry

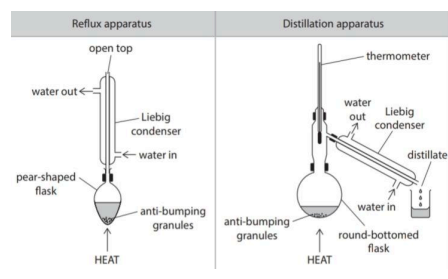
❖ Heat Under Reflux vs Distillation:



P.O.C	Heat under Reflux	Distillation	Heat in a sealed system/ under pressure
Reactions	- Elimination reactions. - Formation of nitriles/alcohols/alkenes from halogenoalkanes. - Bromination and iodination of alcohols. - Oxidation of primary alcohol to carboxylic acid	Oxidation of primary alcohols to aldehydes	Formation of amines from halogenoalkanes

and secondary alcohol to ketone.

Diagram



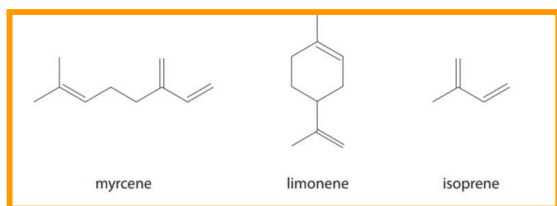
- Give a reason why cyclohexanol and cyclohexanone can be separated by fractional distillation but not by simple distillation.
 - the boiling temperatures are (too) similar/close (to each other)

❖ Geometric Isomers:

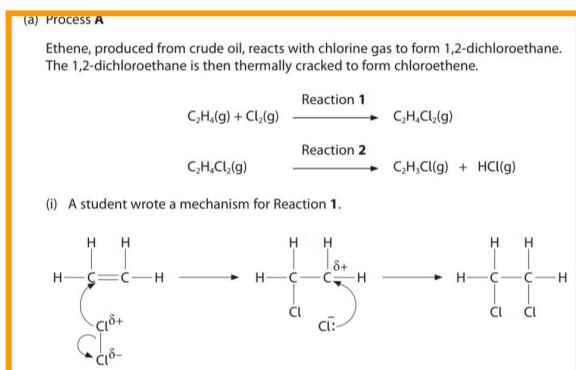
- Explain why Z-hex-3-enal and E-hex-2-enal are structural isomers but not geometric isomers.
 - (structural isomers as) same molecular formula / both $C_6H_{10}O$
 - (not geometric isomers as) the double bond/ $C=C$ is in a different position.
- Explain whether or not these two alkenes (2-methylbut-1-ene and 2-methylbut-2-ene) show geometric isomerism.
 - (in both cases) there are two identical (functional) groups/atoms on one carbon (of the double bond)
 - neither display geometric isomerism

❖ Reactions of Alkenes:

- The process carried out on alkanes obtained from crude oil to increase the octane number of a fuel is reforming/reformation.
- $C=O$ is shorter than $C-O$ bond, has both a sigma and pi bond, and has 4 shared electrons.
- From alkene to alkane, the reaction is addition reaction.
- State the conditions required for hydrogenation of alkenes.
 - heat and nickel / Ni (catalyst)



- Show that myrcene and limonene are terpenes by comparing the molecular and/or empirical formulae of myrcene, limonene and isoprene.
 - molecular formula of either myrcene, limonene or both is $C_{10}H_{16}$ and the molecular / empirical formula of isoprene is C_5H_8 , so the molecular formula of myrcene **and** limonene is twice isoprene (so they are terpenes)
 - OR molecular formula of either myrcene, limonene or both is $C_{10}H_{16}$ and the molecular / empirical formula of isoprene is C_5H_8 , so the empirical formula of myrcene **and** limonene is the same as isoprene (so they are terpenes).
- Iodine monochloride reacts with propene to form two isomeric products, 2-chloro-1-iodopropane, and 1-chloro-2-iodopropane. This is an addition reaction that is similar to the reaction of propene with hydrogen halides. Explain which of these isomers is the major product.
 - 2-chloro-1-iodopropane
 - iodine is partially positive and is attacked by π electrons, more stable secondary carbocation formed.



The mechanism contains two mistakes. Identify each mistake, giving the correction that should be made.

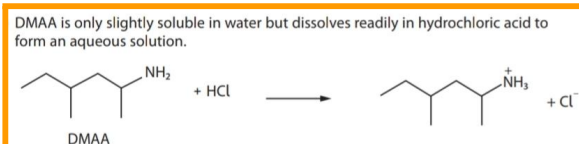
- Mistake 1: the arrow pointing to the $C=C$ bond is incorrect.
- Correction: the arrow should be pointing away from the bond

- Mistake 2: the partial charge on the C in the intermediate is incorrect
- Correction: it should be a full positive charge.

- Suggest suitable reagents and conditions for the conversion of propene to propane-1,2-diol.
 - potassium manganate(VII)/potassium permanganate
 - acidified/cold/room temperature/dilute aqueous solution.

❖ Bond angles, types and Sustainability:

- Give the reason why chlorine dioxide is not a linear molecule.
 - chlorine has non-bonding electrons in its outer shell which repel the bonding pairs (of electrons).
- For H_3O^+ , what shape and H-O-H bond angle are predicted by the electron-pair repulsion theory? (trigonal pyramid, 107°)
- Suggest why the production of propenoic acid from propane-1,2,3-triol is more sustainable than its production from propene.
 - from propene the starting material is crude oil which is non-renewable/finite
 - from propane-1,2,3-triol, the starting material is from biomass/uses a by-product/reduces waste from biodiesel production
 - propane-1,2,3-triol route produces only water as unwanted product
 - from propene manganese compounds need to be separated



Explain the type of bonding that occurs between the nitrogen atom and the hydrogen ion, when the positive ion forms.

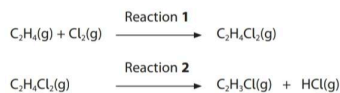
- (the bond formed is a) dative (covalent) / coordinate bond
- As the (lone) pair of electrons on the nitrogen (atom)
- (form the bond) as hydrogen (ion) has an empty orbital / no electrons.

❖ Atom Economy:

- If the atom economy for a reaction is 100% although it produces two products or more, then all the products are desired/ useful products.

(a) Process A

Ethene, produced from crude oil, reacts with chlorine gas to form 1,2-dichloroethane. The 1,2-dichloroethane is then thermally cracked to form chloroethene.



Process B

In an alternative process, chloroethene is obtained by reacting ethyne ($\text{HC}\equiv\text{CH}$), with hydrogen chloride. Ethyne is produced from coal. This reaction uses a catalyst of mercury(II) chloride which is highly toxic.



The temperature during this reaction is high enough to vaporise the mercury(II) chloride catalyst.

State how Process A compares to Process B in terms of the percentage atom economy by mass and environmental impact of each process. No calculation is needed.

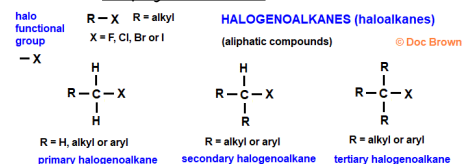
- Atom economy (of process A is < 100% but) in process B it is 100%
- in process A $\text{HCl}(\text{g})$ is produced which is toxic / corrosive OR catalyst for process B / Mercury / Mercury(II) chloride is highly toxic
- Allow **both** processes use non-renewable starting material.

Halogenoalkanes

❖ Classification of Halogenoalkanes & Rate of reactions:

Classification

→ Displayed Formula:



→ Structural Formula:

◆ Primary:



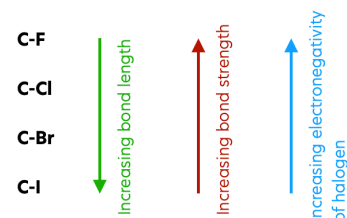
◆ Secondary:



◆ Tertiary:



Carbon-halogen bond



Rate of reaction (e.g. hydrolysis)

- Rate of reaction depends on the strength of the carbon-halogen bond. This strength decreases **down** the group, leading to a **faster rate** and **shorter time** taken for the reaction to be completed, or for a precipitate to appear.
- In terms of fastest rate of reactions:
 - ◆ C-X bond:
 - $\text{C-I} > \text{C-Br} > \text{C-X}$
 - ◆ Classification:
 - $3^\circ > 2^\circ > 1^\circ$
- If same compound, same displayed formula/isomer (e.g. n-chloroalkane, n-bromoalkane, and n-iodoalkane), but different C-X bond:
 - ◆ n-iodoalkane would have the **fastest** rate of reaction (e.g. fastest rate of hydrolysis), since the **C-I bond** is the **weakest**, and has **lowest** bond enthalpy.
- If same compound, same C-X bond, but different isomer (e.g. n-iodo-m-methylalkane and n-iodoalkane):
 - ◆ **tertiary** iodoalkanes would have **highest** rate of reaction.

- State what would be measured in an experiment to compare the rates of hydrolysis of halogenoalkanes.
 - time taken for the (first appearance of the) precipitate (of silver halide) to form.
- State which of these halogenoalkanes (1-chloropropane, 1-bromopropane, 1-iodopropane) would hydrolyse the fastest. Justify your answer.
 - 1-iodopropane, C-I bond weakest/lowest bond enthalpy.
- Samples of 1-iodobutane and 2-chloro-2-methylpropane were hydrolysed using aqueous silver nitrate in ethanol. Explain whether or not it is possible to predict the relative rate of hydrolysis of these two compounds.
 - 2-chloro-2-methylpropane should react faster than 1-iodobutane because it is tertiary
 - 1-iodobutane should react faster than 2-chloro-2-methylpropane because the C-I bond enthalpy is lower than C-Cl
 - it is not possible to predict the relative effects of these two opposing factors / structure and bond enthalpies.

The relative rates of hydrolysis of a series of halogenoalkanes were determined using the following method:

- five test tubes, each containing 2 cm³ of ethanol and 2 cm³ of aqueous silver nitrate, were placed in a water bath at 50°C
- four drops of a different halogenoalkane were added to each test tube
- the time taken for a precipitate to appear in each test tube was recorded.

The results are shown.

Halogenoalkane	Time for ppt to appear / s
2-bromobutane	29
2-chlorobutane	75
2-iodobutane	<1
1-bromobutane	41
2-bromo-2-methylbutane	13

*Explain these results by considering: the chemical reaction occurring, the structures of the halogenoalkanes, and the strengths of the carbon-halogen bonds.

- (mechanism is nucleophilic) substitution
- (precipitation/precipitate of) silver halide forms
- the time taken for the precipitate to form/rate of reaction depends on the carbon – halogen bond strength
- strength/bond enthalpy of the C-Halogen bond decreases going down group 7/any correct comparison of at least two Hal-C

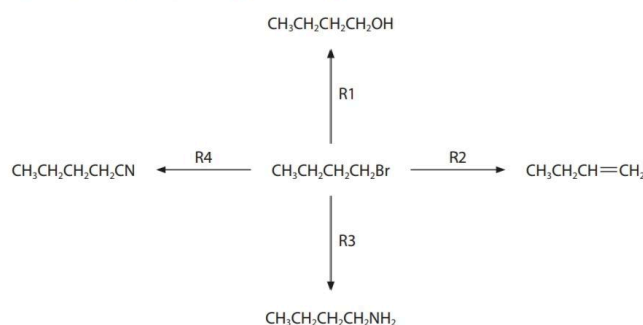
bond strengths (because the atomic radius increases)

- the reactivity/rate of reaction of the bromoalkanes increases $1^{\text{ry}} < 2^{\text{ry}} < 3^{\text{ry}}$

❖ Reactions of Halogenoalkanes:

→ Reactions and Conditions:

(c) Some reactions of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ are shown.

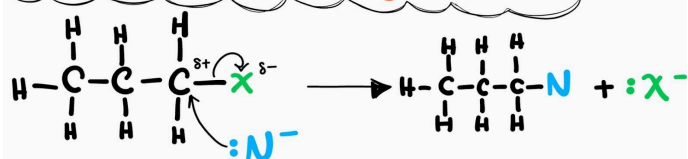


Reaction	Mechanism	Reagent	Ion/molecule type	Conditions
R1	Nucleophilic substitution	Potassium hydroxide	OH ⁻ (water) Nucleophile	Cold, dilute, aqueous KOH (water solvent), heat under reflux
R2	Elimination	Potassium hydroxide	OH ⁻ Base	Hot, conc, ethanolic KOH (ethanolic solvent), heat under reflux
R3	Nucleophilic substitution	(Excess) Ammonia	NH ₃ Nucleophile	Ethanolic NH ₃ (ethanolic solvent), heat under pressure/in a sealed tube
R4	Nucleophilic substitution	Potassium cyanide	CN ⁻ Nucleophile	Ethanolic KCN (ethanolic solvent), heat under reflux

★ Organic compounds, such as C₄H₁₀ and higher alkenes, when produced from elimination of halogenoalkanes, will include structural isomers (-1-ene/-2-ene/-3-ene/...) and geometric isomers (Z/E or Cis/Trans).

→ General mechanism:

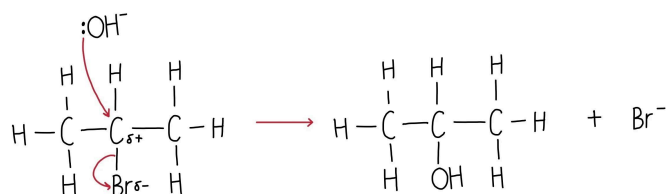
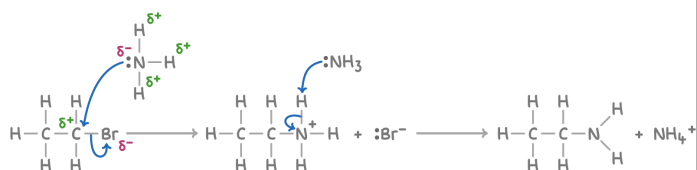
Nucleophilic Substitution of Primary halogenoalkanes



X = halogen

$:N^-$ = nucleophile

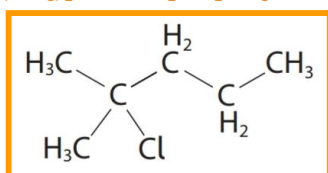
→ Reaction-Specific mechanism:



BROMOPROPANE

PROPANOL

- The chloroalkane with six carbon atoms shown can be converted into an alcohol using potassium hydroxide. Complete the equation for this reaction. State symbols are not required.



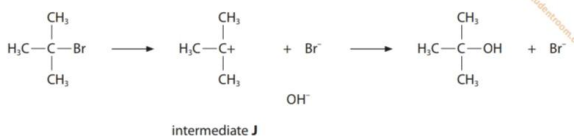
- Explain why two alkenes are formed in the reaction of chloroalkane with 6 carbon atoms with ethanolic solution of KOH.
 - hydrogen may be removed from the 1-position or 3-position OR hydrogen may be removed from either side of the carbon bonded to chlorine
 - forming a 1-ene or a 2-ene OR forming the double bond between the first and second carbons or the second and third carbons OR forming 2-methylpent-1-ene or 2-methylpent-2-ene.

- Give the reason why chemists often use the reaction of formation of a nitrile, when synthesising organic compounds.
 - to increase the length of a **carbon chain** / to extend the **carbon chain**.
- Further substitution of the ethylamine molecule may occur if the halogenoalkane is in excess. State two features of the ethylamine molecule that make this possible.
 - (ethylamine) has a lone pair on the nitrogen (and a) hydrogen(s) attached to the nitrogen (which can be substituted)
- (R.Q.): *Discuss the reactions between sodium hydroxide and 2-chlorobutane. The hydroxide ion acts in two different ways depending on the reaction conditions. Include the structures of four possible products, the reaction conditions needed and the roles of the hydroxide ion.
 - both reactions involve hydroxide ions / OH^- , but in aqueous conditions/reaction with aqueous solution is substitution (to produce) the alcohol, $CH_3CHOHCH_2CH_3$ (butan-2-ol), whereas, in ethanolic/alcoholic conditions/reaction with ethanolic/alcoholic solution is elimination (to produce) the alkene, $CH_2=CHCH_2CH_3$ (but-1-ene), $E-CH_3CH=CHCH_3$ (E-but-2-ene), and $Z-CH_3CH=CHCH_3$ (Z-but-2-ene) is formed.

Skeletal formula	Name
	Z-but-2-ene
	E-but-2-ene
	but-1-ene

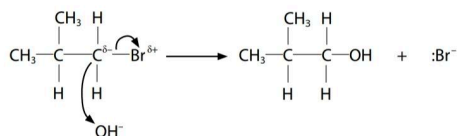
- in substitution/aqueous solution, the hydroxide ion acts as a nucleophile **and** in elimination/ethanolic solution, the hydroxide ion acts as a base.
- the best solvent for a halogenoalkane is an alcohol.
- Halogenoalkanes have higher BP than alkanes.
- Number of structural isomers of C_3H_6BrI is 5.

(d) An incomplete mechanism for Reaction 2 is shown.



- Describe how intermediate J is formed in the first step, naming the process that occurs.
 - heterolytic (bond) fission occurs
 - the (pair of) electrons in the C—Br bond are transferred to the bromine atom
- $\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}_2\text{CH}_3$ is heated with ethanolic potassium hydroxide. How many **alkene** products are possible? 4.
- Suggest why the yield of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ obtained from reacting ammonia with 1-bromopropane is low.
 - secondary amine / tertiary amine / quaternary (ammonium) salt may form
- Give a reason why this reaction (formation of amines) should be carried out by heating the reactants in a sealed tube rather than heating under reflux.
 - ammonia (gas) would escape (from the condenser when heated under reflux) OR to prevent ammonia (gas) escaping (from the condenser when heated under reflux)

1-bromo-2-methylpropane can be converted back to 2-methylpropan-1-ol by heating with aqueous alkali. A student suggested the following mechanism for the reaction.



- Identify and correct the three mistakes in the mechanism shown.
 - C-Br dipole reversed
 - OH^- to C arrow reversed
 - lone pair missing (from OH^-)

Alcohols

❖ Classification of alcohols:

Displayed	Classification of Alcohols
	Primary alcohol Ethanol Secondary alcohol Isopropanol Tertiary alcohol tert-Butanol
Structural	→ <u>Primary alcohol:</u> RCH_2OH → <u>Secondary alcohol:</u> RCHOH → <u>Tertiary alcohol:</u> RCOH

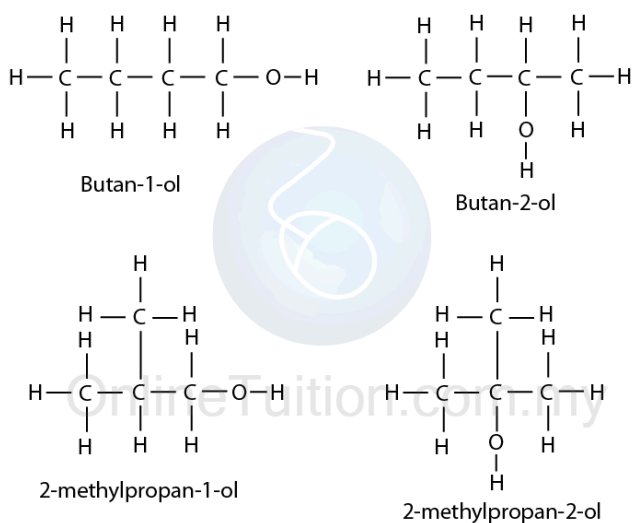
Term	Definition
Alcohols	contains the OH group / OH is the alcohol group
primary alcohols	primary means the C bonded to the OH (group) is attached to 1 (or 0) alkyl group / carbon atom / R group
secondary alcohol	the carbon (atom) attached to the OH group is attached to two other carbon (atoms) / alkyl groups.

- Saturated non-cyclic alcohols with six carbon atoms can be classified as one of three types. Explain these three types of alcohols, giving an example of each.
 - alcohols are primary, secondary or tertiary
 - a primary alcohol has two (or three) hydrogens attached to the carbon with the —OH attached OR a primary alcohol has one (or no) carbon/R group attached to the carbon with the —OH attached **and** a secondary alcohol has one hydrogen attached to the carbon with the —OH attached OR a secondary alcohol has two

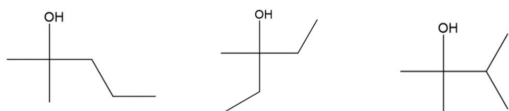
carbon/R groups attached to the carbon with the $-OH$ attached **and** a tertiary alcohol has no hydrogens attached to the carbon with the $-OH$ attached OR a tertiary alcohol has three carbon/R groups attached to the carbon with the $-OH$ attached

- example of any primary alcohol and any secondary alcohol and any tertiary alcohol, all examples of alcohols are saturated and non-cyclic and contain six carbon atoms.

Isomers of Butanol:



- Draw the skeletal formula of each of the three tertiary alcohols with the formula $C_6H_{14}O$.



Name	Butan-1-ol	butan-2-ol	(2-)methylpropan-2-ol
Displayed formula	$\begin{array}{ccccccc} & H & H & H & H & & \\ & & & & & & \\ H & -C & -C & -C & -C & -O-H \\ & & & & & & \\ & H & H & H & H & & \end{array}$	$\begin{array}{ccccccc} & H & H & OH & H & & \\ & & & & & & \\ H & -C & -C & -C & -C & -H \\ & & & & & & \\ & H & H & H & H & & \end{array}$	$\begin{array}{ccccccc} & H & & & & & \\ & & & & & & \\ H & -C & -H & -C & -H \\ & & & & & & \\ & H & & O & & & \\ & & & & & & \\ & & & H & & & \end{array}$
Classification of alcohol	primary	secondary	tertiary

Skeletal formula	Name	Classification	Boiling temperature / °C
	butan-1-ol	primary	117
	(2-)methylpropan-1-ol	primary	108
	butan-2-ol	secondary	99
	2-methylpropan-2-ol	tertiary	82

→ The more OH gps, the higher the bp.

❖ Oxidation of Alcohols:

A GUIDE TO OXIDATION REACTIONS OF ALCOHOLS

Compounds containing the alcohol functional group ($-OH$) can be oxidised to produce carbonyl compounds. Here's how it happens.

THE REAGENTS

Primary (1°) alcohols: H_3C-CH_2-OH
Secondary (2°) alcohols: $H_3C-CH(OH)-CH_3$
Tertiary (3°) alcohols: $H_3C-C(OH)(CH_3)_2$

Alcohols can be oxidised to carbonyl compounds (containing a $C=O$ bond) using an oxidising agent. Acidified dichromate (VI) salts can be used, though due to their toxicity alternative reagents can also be utilised, such as pyridinium chlorochromate (PCC).

SODIUM DICHROMATE $Na_2Cr_2O_7$
POTASSIUM DICHROMATE $K_2Cr_2O_7$

The type of compound obtained from the reaction depends on the starting alcohol (shown below). When an oxidation reaction is carried out with a dichromate salt, the dichromate ion ($Cr_2O_7^{2-}$) is reduced to the Cr^{3+} ion, giving a colour change from orange to dark green.

1° ALCOHOL → ALDEHYDE
2° ALCOHOL → KETONE
3° ALCOHOL → NO REACTION

PRODUCTS WITH DIFFERENT ALCOHOLS

THE APPARATUS

Distillation: A round-bottom flask containing the alcohol and reagents is heated. The mixture is distilled into a Liebig condenser, which is cooled by water. The product is collected in a receiver flask.

Heating under reflux: A round-bottom flask containing the alcohol and reagents is heated. The mixture is refluxed in a vertical Liebig condenser, which is cooled by water. The product is collected in a receiver flask.

An aldehyde can be obtained from primary alcohols using distillation (above). Otherwise, heating under reflux (below) is used to make sure the alcohol is fully oxidised before distilling off the product.

TESTING FOR REACTION PRODUCTS

Oxidising agents can be represented simply in chemical equations as $[O]$. An example reaction is shown below.

ETHANOL $\xrightarrow{[O]}$ ETHANAL $\xrightarrow{[O]}$ ETHANOIC ACID

Reaction in step 1: $3 C_2H_5OH + 2 Cr_2O_7^{2-} + 16 H^+ \rightarrow 3 CH_3COOH + 4 Cr^{3+} + 11 H_2O$

There are two different chemical reactions that can be used to identify the products of oxidation reactions.

FETTLING'S SOLUTION: Contains complexed Cu^2 ions. Aldehydes reduce these ions to red copper(I) oxide, forming a red precipitate. Ketones do not react with Fehling's solution.

TOLLEN'S REAGENT: Contains the diamine silver ion, $[Ag(NH_3)_2]^+$. Aldehydes reduce this ion to metallic silver, forming a silver mirror on the glass surface. Ketones do not react with Tollen's reagent.

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- Describe in outline the technique for converting an alcohol such as Z-hex-3-en-1-ol into an aldehyde such as Z-hex-3-enal in a laboratory. You may include a diagram.
 - heat with potassium dichromate(VI)/ $K_2Cr_2O_7$ and sulfuric acid/ H_2SO_4
 - distil, so (aldehyde) is not oxidised further/does not form a carboxylic acid.
- Give the reagents and conditions that are needed for the conversion of alcohol to aldehyde in the laboratory. Include in your answer any colour change seen.
 - Reagents:
 - sulfuric acid / H_2SO_4
 - potassium dichromate(VI) / $K_2Cr_2O_7$
 - Conditions:
 - distillation
 - Colour change:
 - orange to dark green

■ From orange to green.

- Cyclohexanol can be converted to cyclohexanone. Name the reagents and conditions for this conversion.

- sodium/potassium dichromate(VI)/ $K_2Cr_2O_7$ / $Na_2Cr_2O_7$ and (dilute) sulfuric acid/ H_2SO_4
- (heat under) reflux.

- Methanol can be oxidised to produce methanoic acid. State the reagents and reaction conditions for this oxidation in the laboratory, include an equation.

- Reagents

- acidified / H^+ / sulfuric acid sodium/potassium dichromate(VI)/ $K_2Cr_2O_7$ / $Na_2Cr_2O_7$ / $Cr_2O_7^{2-}$

- Reaction conditions

- (Heat under) reflux.

- Equation



- State the conditions needed to maximise the yield of each product from the oxidation of alcohol.

- Conditions for maximum yield of ethanal:

- Distil (off the ethanal from reaction mixture as it forms)

- Conditions for maximum yield of ethanoic acid:

- (heat under) reflux

- Describe how the reflux apparatus ensures that any ethanal initially produced is further oxidised to ethanoic acid.

- any ethanal (that evaporates) is condensed back into the flask to be further oxidised (to ethanoic acid).

- *Propan-1-ol is heated with a concentrated solution of acidified potassium dichromate(VI). Explain how the conditions used affect the rate of the reaction and ensure that propanoic acid is the only organic product.

- The higher the concentration (of acid or $Cr_2O_7^{2-}$) the higher the rate $\rightarrow$ because the collision frequency increases.
- The higher the temperature the faster the rate $\rightarrow$ because more particles have an energy greater than the activation energy / more successful collisions.

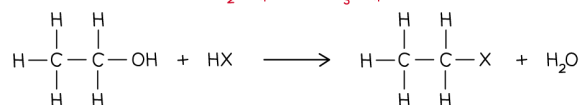
- excess / concentrated oxidising agent and heat under reflux ensures complete oxidation.

- If an alcohol reacts with acidified potassium dichromate(VI) solution to form two different organic products, depending on the reaction conditions, then it is a primary alcohol.

❖ Tests for Presence of -OH group:

SUBSTITUTION OF ALCOHOLS

- HX (OR KBr WITH H_2SO_4 OR H_3PO_4)



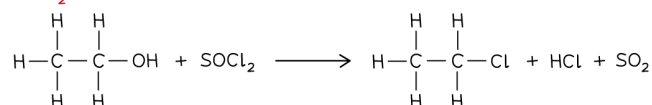
- PCl_3 + HEAT



- PCl_5



- $SOCl_2$



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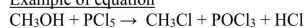
- Describe a chemical test and its positive result for the alcohol functional group in alcohol Y.

- add PCl_5 / phosphorus(V) chloride /phosphorus pentachloride $\rightarrow$ (white) misty/steamy fumes (of HCl) evolved (that turn damp blue litmus red / form white smoke with ammonia)
- OR potassium dichromate(VI)/ $K_2Cr_2O_7$ and sulfuric acid/ H_2SO_4 $\rightarrow$ colour change from orange to green.
- OR Add sodium / Na $\rightarrow$ Effervescence / bubbles seen / fizzing
- OR Add Lucas' Reagent $\rightarrow$ Solution turns cloudy immediately / quickly

Reaction between methanol and phosphorus(V) chloride

Equation	Observations
----------	--------------

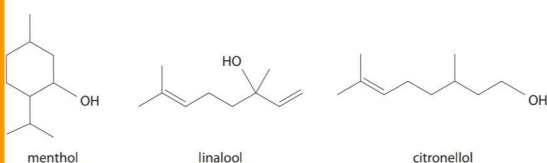
Example of equation



steamy/ misty/ white fumes

- Acetoin contains two functional groups. Some chemical tests were carried out on acetoin. These tests identify one of the two functional groups. Acetoin produced steamy fumes when reacted with PCl_5 . Acetoin did not react with sodium hydrogencarbonate solution. Acetoin turned hot acidified potassium dichromate(VI) solution from orange to green. State what can be deduced from each of these three tests and hence identify this functional group.
 - (acetoin reacted with PCl_5) so must contain an OH group
 - (acetoin did not react with sodium hydrogencarbonate so) it is not a carboxylic acid / it is an alcohol
 - (acetoin reacted with potassium dichromate(VI) solution so must) contain a 1^{ry} or 2^{ry} alcohol.

Three students were asked to carry out a single chemical test to unambiguously identify the presence of the –OH group in the three terpenoid compounds, menthol, linalool and citronellol.



The first student tried using acidified potassium dichromate(VI) to oxidise the alcohols.

The second tried acidified potassium manganate(VII), an oxidising agent similar to potassium dichromate(VI).

The third student tried using phosphorus(V) chloride.

*Critique the three students' suggestions by linking the expected results, and the actual observations that would be made, for each test and the structures of the three alcohols.

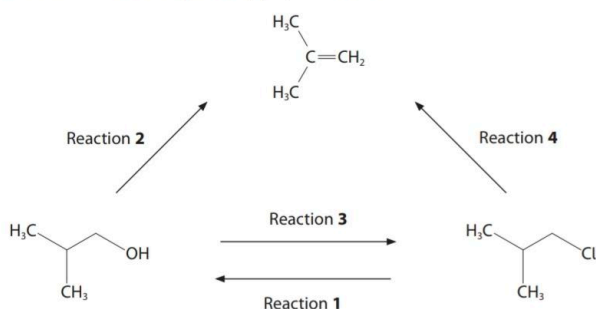
- (potassium dichromate(VI) is not a good choice because)
 - potassium dichromate(VI) would go from **orange to green** (with menthol and citronellol but would stay the same / would not go green with linalool)
 - menthol and citronellol are oxidised / react but linalool is a tertiary alcohol which is not oxidised by / does not react with potassium dichromate (so does not identify this –OH group)

- (potassium manganate(VII) is not a good choice because)
 - potassium manganate(VII) would go from **purple to colourless** (with all three alcohols)
 - (but would not identify the presence of the –OH group) as it also reacts with the C=C / reacts to produce a diol with linalool and citronellol / two of the alcohols
- (Phosphorus(V) chloride is the best choice because)
 - phosphorus(V) chloride will produce **misty fumes** (with all)
 - (and identifies the presence of the –OH group in each as it) reacts with primary secondary and tertiary alcohols / reacts with all three alcohols

❖ Reactions of Alcohols:

- Explain why anhydrous calcium chloride is added to ethanol before it is used to dissolve the sodium hydroxide.
 - to remove water (from the ethanol) , as it is a drying agent.
 - to prevent (nucleophilic) substitution from taking place.
- Identify, by name or formula, a suitable drying agent.
 - (Anhydrous) sodium sulfate / Na_2SO_4 / magnesium sulfate / MgSO_4 / calcium chloride / CaCl_2 / calcium sulfate / CaSO_4 / calcium oxide / CaO .
- Identify the reagent or reagents you would use to make 1-bromopropane from propan-1-ol.
 - 50% / concentrated sulfuric acid / H_2SO_4 and potassium bromide / KBr .
- 2-methylpropan-1-ol can be converted to 1-bromo-2-methylpropane. Give the reagents and conditions used for this reaction.
 - KBr /potassium bromide and (50%) sulfuric acid
 - (heat under) reflux
- Which reagent(s) could be used to produce a chloroalkane from a tertiary alcohol? PCl_5 , concentrated HCl , concentrated H_2SO_4 and KCl .

(d) A reaction scheme involving 2-methylpropene is shown.



Reaction Number	Reagents, conditions	Reaction type
R2	reagent: (concentrated) phosphoric(V) acid/ H_3PO_4 Conditions: 300°C, 60 atm, alumina catalyst	Elimination reaction
R3	reagent: phosphorus(V) chloride/ phosphorus pentachloride/ PCl_5 Conditions: room temperature	Nucleophilic substitution

❖ Other Organic Tests:

Test	Observations	
	w/Aldehyde	w/Ketone
Warm with Fehling's solution	Red precipitate	No change/reaction
add solid phosphorus(V) chloride	No change/reaction	
warm with acidified potassium dichromate(VI)	Turns green	No change/reaction
add sodium hydrogencarbonate	No change/reaction	
Add Tollen's reagent	Silver mirror	No change/reaction
Iodine + alkali (NaOH) [Iodoform test]	No change/reaction	Yellow ppt. of Iodoform

Reaction Occurs?	Alcohol	Aldehyde
adding a very small piece of sodium metal to each liquid	✓	✗

adding sodium carbonate solution to each liquid	✓	✓
warming each liquid with Fehling's solution	✗	✓
adding PCl_5 to each liquid	✓	✗

- Give a chemical test, and its positive result, that could be used to confirm the functional group in methanoic acid.
 - Functional group test
 - add any carbonate /hydrogencarbonate OR add magnesium (powder/ribbon)
 - OR addition of any alcohol (and strong acid to form an ester)
 - Positive result
 - effervescence / bubbles / gas given off (turns lime water cloudy OR burns with a "pop")
 - OR Fruity smell
- A test to confirm the presence of an aqueous acid is adding a small amount of solid sodium carbonate to the solution. Describe two observations you would see in this test.
 - (white) solid / crystals / sodium carbonate dissolves
 - (colourless) bubbles(of gas) / effervescence / fizzing.
- A student suggested using universal indicator to check for the presence of propanoic acid formed in the reaction mixture. Give a reason why the result of this test is **not** likely to be conclusive.
 - colour of the potassium dichromate(VI) / chromium(III) will mask the colour of the indicator
 - OR the reaction mixture will contain hydrogen ions / acid (present from the oxidising agent).

Reagent	Observation	Functional group
bromine water	bromine water decolourised	$\text{C}=\text{C}$
aqueous sodium carbonate	effervescence	$-\text{COOH}$

- Adding sodium carbonate solution does **NOT** distinguish between ethanol and ethanal.
- How many structural isomers with the molecular formula $C_5H_{10}O$ react with Benedict's or Fehling's solutions? 4.
- Give two reasons why this reaction (adding PCl_5 to alcohol) would produce a greater yield of this chloroalkane than the combined reactions in (b) (elimination reaction of alcohol) and © (electrophilic addition of hydrogen chloride and alkene)
 - *this reaction is one step / the other method is a two-step process*
 - *yield is higher because in the 2 step process both steps give 2 isomers/this reaction has no isomers /no organic by-products /no minor product.*
- Suggest two possible advantages, other than increasing the yield, of a one-step synthesis reaction.
 - *(a catalyst) may reduce the operating temperature*
 - *a lower pressure may be used which is safer / requires less expensive/specialised equipment*
 - *the reaction may not proceed via toxic gas / greenhouse gases and/or may go to completion (rather than being in equilibrium)*
 - *only one reaction vessel required*
 - *may produce fewer by-products (or no isomers if it's an organic compound), take less time, and/or require less energy if a lower pressure is used.*

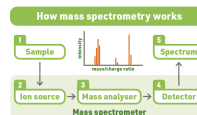
Mass and IR Spectra

❖ Mass Spectra:

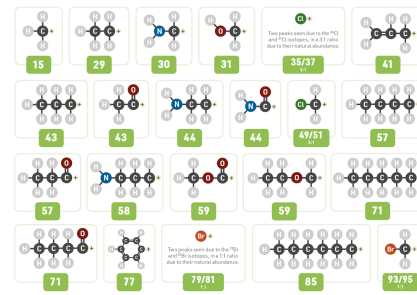
A guide to interpreting mass spectra



Mass spectrometry is an analytical technique that allows us to measure the masses of atoms and molecules. The most important peak in a mass spectrum is the molecular ion peak, which can be used to determine the mass of the molecule, but fragment ions can also provide information on chemical structure.



- 1 A small sample of the substance to be analysed is added to the mass spectrometer.
- 2 The mass spectrometer ionises the sample. This can be done in a number of ways, including with a laser, applying a voltage to a liquid sample spray, or firing electrons at a gaseous sample. Some molecules fragment into smaller ions.
- 3 A mass analyser separates ions based on their mass/charge ratio. This can be done in a number of ways using an electric and/or magnetic field.
- 4 Ions hit the detector and it converts them into a signal, amplifies it, and records it.
- 5 The signal is output as a mass spectrum.



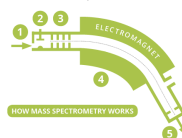
A selection of common fragment ions seen in mass spectra are shown above, along with their masses. Note that the structures shown are general representations, and it can also be possible for isomeric structures (those with the same constituent atoms, but a different structure) to cause the peaks in spectra. There are many more fragments possible than those shown, but knowledge of these fragments should suffice to interpret spectra of most simple molecules.

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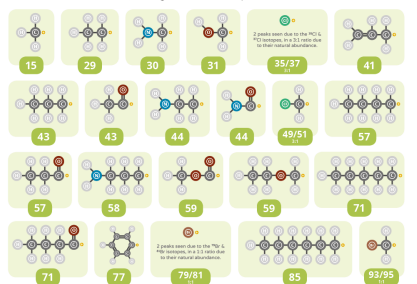
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A GUIDE TO INTERPRETING MASS SPECTRA

Mass spectrometry is an analytical technique that allows us to measure the masses of atoms and molecules. The most important peak in a mass spectrum is the molecular ion peak, which can be used to determine the mass of the molecule, but fragment ions can also provide information on chemical structure.



- 1 The sample is introduced to the mass spectrometer. Only very small samples are required. A heater is often present to evaporate the sample.
- 2 An electron gun ionises molecules in the sample by knocking out electrons, producing positive ions. Some molecules break into smaller ions & fragments.
- 3 The positive ions generated are passed through an electric field which accelerates them into a magnetic field generated by an electromagnet.
- 4 As the positive ions pass through the magnetic field, they are deflected. Lighter ions are deflected more than heavier ions, as are those with higher charges.
- 5 The positive ions hit a charged plate & accept electrons, forming a signal. The more ions that hit, the greater the signal. The output is a mass spectrum.

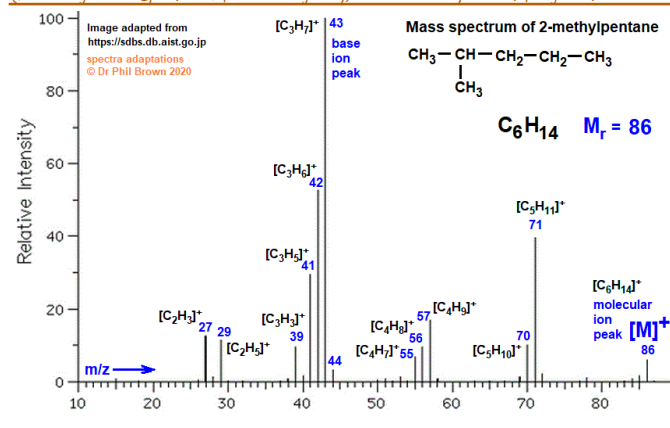


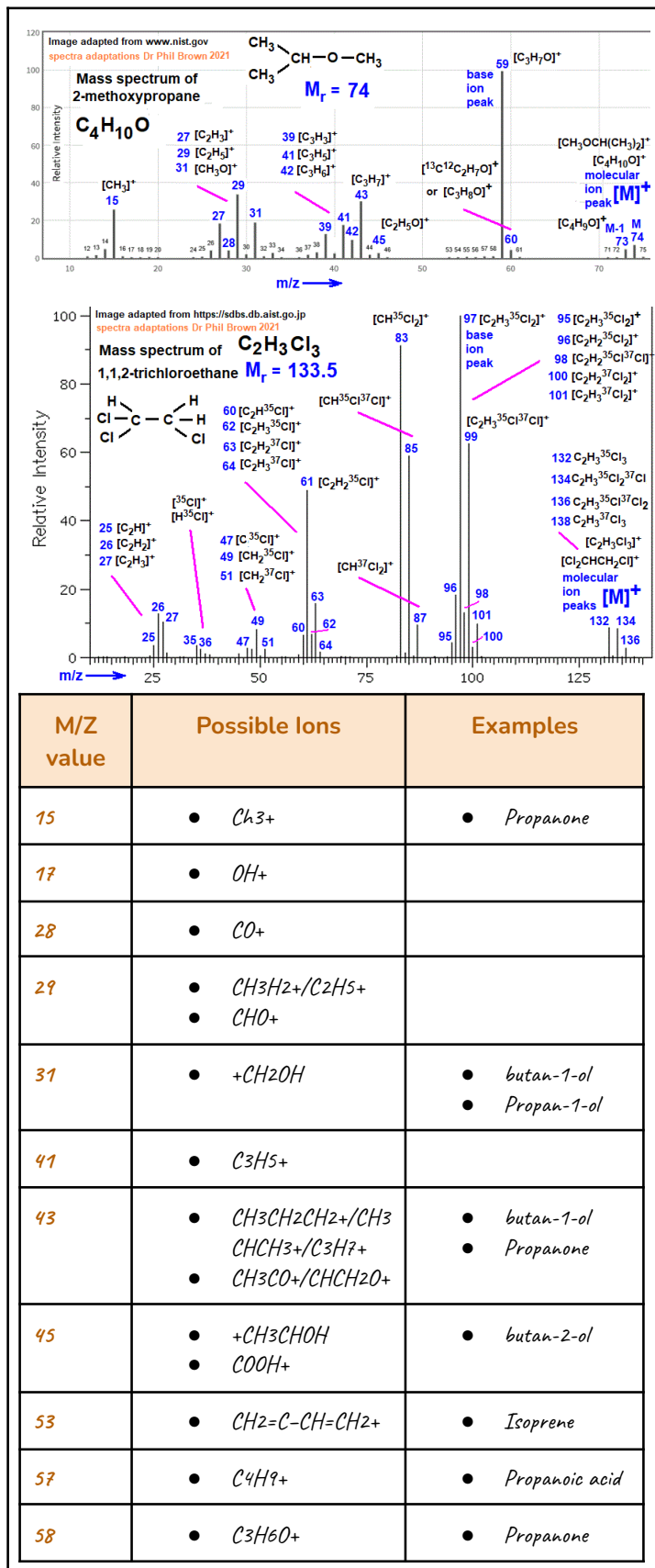
Above are shown a selection of common fragment ions seen in mass spectra, along with their masses. Note that the structures shown are general representations, and it can also be possible for isomeric structures (those with the same constituent atoms, but a different structure) to cause the peaks in spectra. There are also many more fragments possible than those shown, but knowledge of these should suffice to interpret spectra of most simple molecules.

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(Following mass Spectra is for reading only, but it has important fragmentation ions)

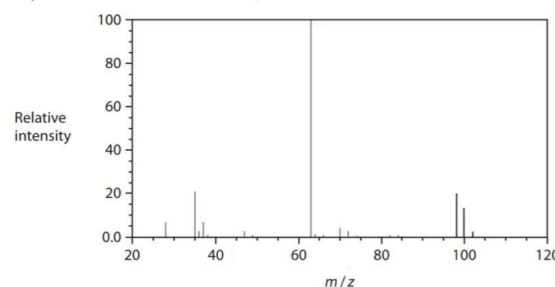




59	• $+(CH_3)_2COH$	• 2-methylpropan-2-ol
68	• $CH_2=C(CH_3)-CH=CH_2^+$	• Isoprene
71	• $C_5H_{11}^+$	
85	• $C_6H_{13}^+$	

- In a mass spectrum, the molecular ion is the ion which always has the highest mass/charge ratio.
- A sample of 1,2-dichloroethane, CH_2ClCH_2Cl , contains only the isotopes 1H , ^{12}C , ^{35}Cl and ^{37}Cl . How many molecular ion peaks are there in its mass spectrum? (3).
- Identify one feature common to the mass spectra of compounds A, B and C which shows that the molecular formula is $C_4H_{10}O$.
 - (all show) parent/molecular ion peak at 74.

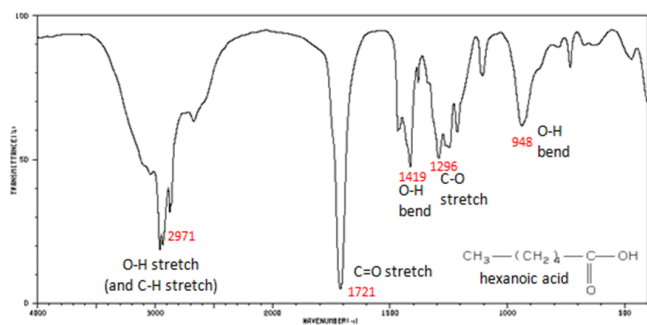
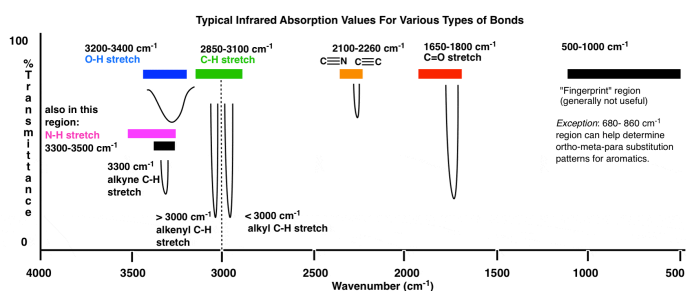
The mass spectrum of a sample of phosgene is shown.
 The peak at $m/z = 65$ has been omitted.



- Give the reason for the ratio of peak heights at m/z values of 102, 100 and 98.
 - chlorine isotopes (35 and 37) are in the ratio of 3:1
 - (as there are) two chlorine atoms give the ratio of 9:6:1

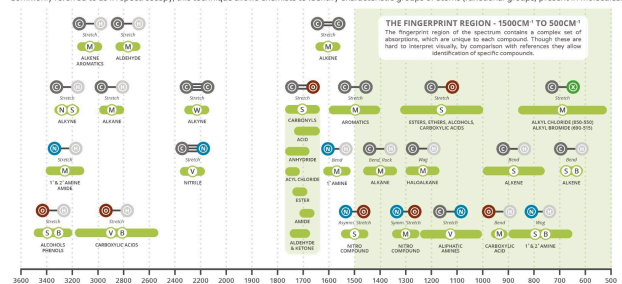
❖ IR Spectra:

WAVENUMBER / cm^{-1}	BOND	FUNCTIONAL GROUP
3750-3200	O-H	alcohol
3500-3300	N-H	amine
3300-2500	O-H	carboxylic acid
3095-3010	C-H	alkene
2962-2853	C-H	alkane
2900-2820	C-H	aldehyde
2775-2700	C-H	aldehyde
1740-1720	C=O	aldehyde
1725-1700	C=O	carboxylic acid
1720-1700	C=O	ketone
1669-1645	C=C	alkene



ANALYTICAL CHEMISTRY - INFRARED SPECTROSCOPY

Commonly referred to as IR spectroscopy, this technique allows chemists to identify characteristic groups of atoms (functional groups) present in molecules.

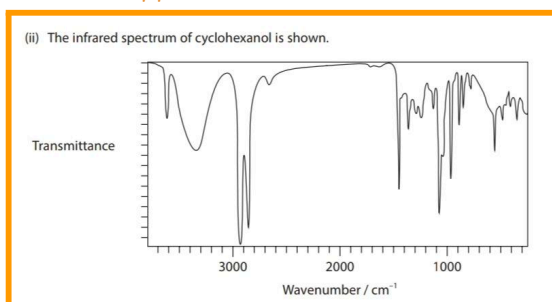


Infrared frequencies make up a portion of the electromagnetic spectrum. If a range of infrared frequencies are shown through an organic compound, some of the frequencies are absorbed by the chemical bonds within the compound. Different chemical bonds absorb different frequencies of infrared radiation. There are a number of characteristic absorptions which allow functional groups (the parts of a compound which give it its particular reactivity) to be identified. This graphic shows a number of these absorptions.

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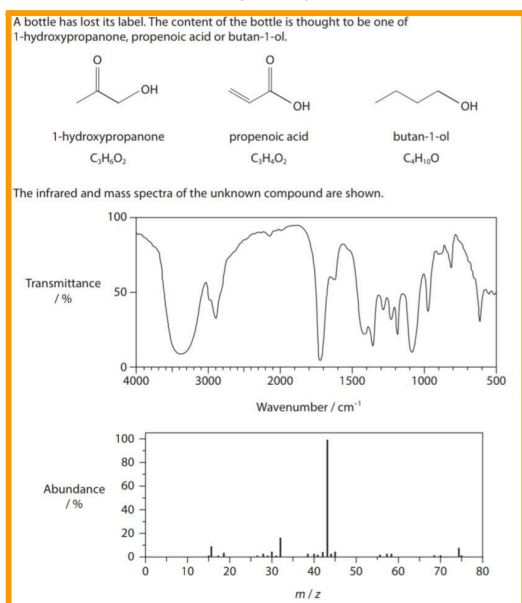
- Identify one feature, common to all three infrared spectra, which shows that A, B and C are all alcohols.
 - peak in the range 3750 – 3200 cm^{-1} and O-H (stretching) bond in alcohols
- State, giving a reason for your answer, if it is possible to identify each of these three alcohols on the basis of the infrared spectra alone.
 - not possible - All three contain the same bonds
 - OR possible - the fingerprint regions differ/by comparing the spectra to reference spectra.
- Describe how the IR spectrum of the reaction mixture would change during this reaction (oxidation of propan-2-ol in acidified $\text{Cr}_2\text{O}_7^{2-}$ ions by heating under reflux).
 - peak at 3750-3200 (cm^{-1} O-H) would diminish, and peak at 1720-1700 (cm^{-1} C=O) would appear.
 - fingerprint region would change (shape) OR peak due to C-O would diminish.
- Explain how the infrared spectra of methanol and methanal can be used to distinguish between the two compounds, stating the relevant bond stretching vibrations and their wavenumber ranges.
 - (methanol) O-H stretch wavenumber range 3750-3200 (cm^{-1})
 - (methanal) C=O stretch wavenumber range 1740-1720 (cm^{-1}) or (methanal) C-H stretch wavenumber range 2900-2820 (cm^{-1}) or (methanal) C-H stretch wavenumber range 2775-2700 (cm^{-1})
- In an experiment, a sample of 2-methylpropan-1-ol was treated with an oxidising agent. Show that infrared spectroscopy can be used to show that both the alcohol and the aldehyde are present in the resulting mixture.
 - (resulting mixture) would give a peak due to O-H bond in alcohols at 3750-3200 (cm^{-1})
 - would give a peak due to C=O bond in aldehydes at 1740-1720 (cm^{-1}) OR would give a peak due to C-H bond in CHO at 2900-2820 and/or 2775-2700 (cm^{-1})

- Explain how the infrared spectrum of methanol and methanoic acid could be used to show that all the methanol has been converted to methanoic acid, quoting relevant bonds and wavenumbers.
 - infra red absorption at 3750 – 3200 cm⁻¹ / 3300 cm⁻¹ due to O-H of alcohol/methanol*
 - infra red absorption at 1725 - 1700 cm⁻¹ / 1700 cm⁻¹ due to C=O group (of carboxylic acid/methanoic acid)*
 - peaks due to methanol/alcohol would disappear/be absent*



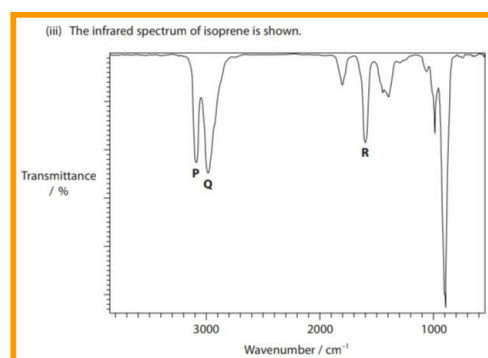
Describe two differences, other than the fingerprint region, that would be expected in the infrared spectrum of cyclohexanone.

- the O-H (stretching absorption) would be absent **and** at wavenumber range 3750 - 3200 (cm⁻¹)*
- the C=O (stretching) absorption would be present **and** at wavenumber range 1720-1700 (cm⁻¹)*



*Compare and contrast the spectra shown with the expected spectra of the three possible compounds, and so identify the unknown.

- O-H bond (broad) absorption at ~3400 (cm⁻¹)*
 - C=O absorption at 1740-1700 (cm⁻¹) so cannot be butan-1-ol*
 - M⁺ has an m/z value of 74 so could be 1-hydroxypropanone or butan-1-ol (propenoic acid has a M⁺ value of 72)*
 - base / most abundant peak has an m/z value of 43 so could be CH₃CO(+) / C₃H₇(+)*
 - no C=C absorption on IR spectra at 1669-1645 (cm⁻¹) OR O-H peak absorbance too high for acid (so not propenoic acid)*
 - so the substance is 1-hydroxypropanone, (not propenoic acid or butan-1-ol).*
- Describe, by referring to wavenumbers and bonds, how the infrared spectra of the isomers produced from elimination reaction differ from the infrared spectrum of alcohol Y.
 - Peak at 3750 – 3200 (cm⁻¹) due to O-H bond present in reactant / absent in product*
 - Peak at 1000-1300 (cm⁻¹) due to C-O bond present in reactant / absent in product*
 - Peak at 1669 – 1645 (cm⁻¹) due to C=C bond present in product / absent in reactant*
 - Peak at 3095 – 3010 (cm⁻¹) due to C-H bond present in alkene in product / absent in reactant.*



State the origins of the peaks labelled P, A, and R.

- P is due to C-H (stretching vibrations) in an alkene.*

- Q is due to C-H (stretching vibrations) in an alkane.
- R is due to C=C (stretching vibrations in an alkene)

Moles and Formulas

$$\text{MOLES} = \text{CONCENTRATION} \times \text{VOLUME}$$

$$\text{MOLES} = \frac{\text{MASS}}{M_r}$$

$$PV = nRT$$

PRESSURE (Pa) →
 VOLUME (m³) →
 MOLES →
 GAS CONSTANT (8.314 J K⁻¹ mol⁻¹) →
 TEMPERATURE (K) →

$$\text{PERCENTAGE YIELD} = \frac{\text{ACTUAL YIELD}}{\text{THEORETICAL YIELD}} \times 100$$

$$\text{ATOM ECONOMY} = \frac{M_r \text{ OF DESIRED PRODUCT}}{M_r \text{ OF TOTAL PRODUCTS}} \times 100$$

$$\text{Percentage} = \left(\frac{\text{Part}}{\text{Total}} \right) \times 100$$

Percentage Uncertainty & Percentage Error

$$\text{Percentage uncertainty} = \frac{\text{absolute uncertainty}}{\text{measured value}} \times 100\%$$

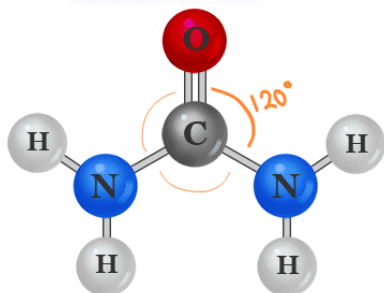
$$\text{Percentage error} = \frac{\text{accepted value} - \text{experimental value}}{\text{accepted value}} \times 100\%$$

APPARATUS	CAPACITY	UNCERTAINTY	PERCENTAGE UNCERTAINTY
burette	50 cm ³	±0.05 cm ³	Note that two burette values are read in a titration, so the total measurement uncertainty is ±0.10 cm ³ . If the titre is 22.50 cm ³ , then the percentage uncertainty is: ±0.10 × 100/22.50 = ±0.44%
pipette	25 cm ³	±0.06 cm ³	The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.06 × 100/25 = ±0.24%
volumetric flask	250 cm ³	±0.3 cm ³	The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.3 × 100/250 = ±0.12%

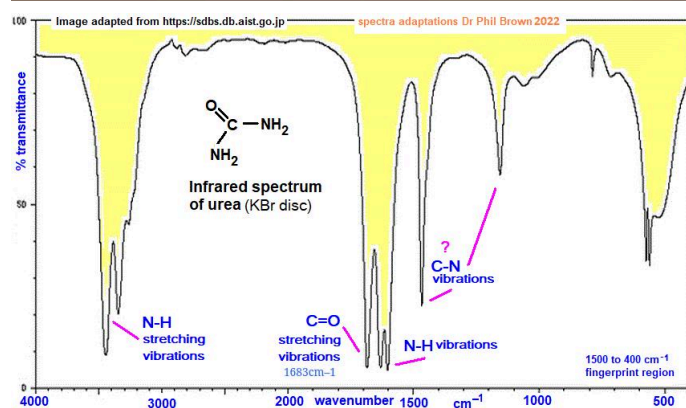
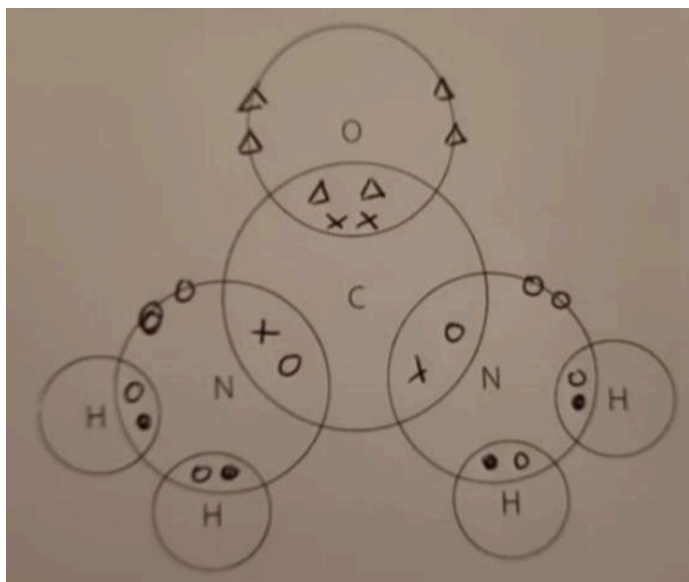
Section C

❖ Urea & Ammonia:

STRUCTURE OF UREA $\text{CO}(\text{NH}_2)_2$



Bond shape: Trigonal Planar



- Give one advantage and one disadvantage of applying ammonia directly into the soil as a fertiliser.
 - advantage: ammonia has a high % N
 - disadvantage: ammonia is a gas (so hard to apply / will escape / will evaporate /

harder to transport/store) / ammonia is a base/alkali/will increase the pH / it is toxic/corrosive.

- Ammonium nitrate, NH_4NO_3 , can be made by reacting ammonia with nitric acid. Give the equation for this reaction, and name the type of reaction occurring.
 - Equation: $\text{HNO}_3 + \text{NH}_3 \rightarrow \text{NH}_4\text{NO}_3$ Or $\text{HNO}_3 + \text{NH}_4\text{OH} \rightarrow \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$.
 - Type: neutralisation (reaction).
- Suggest why urea is used as a fertiliser for crops such as rice but not in regions with unpredictable rainfall patterns.
 - (It is used for crops such as rice because) the ammonia produced will dissolve/react in the water/form ammonium ions (and not be lost)
 - (In areas of unpredictable rainfall the urea will breakdown/decompose and) ammonia will escape/vapourise OR farmers don't know when to apply the urea as it may be wasted.
- Suggest two reasons why it is more profitable to carry out all four stages at the same site, instead of using different sites for each stage in the industrial production of ammonium nitrate.
 - cheaper to produce $\text{H}_2/\text{NH}_3/\text{NO}/\text{HNO}_3$ than to purchase (from other suppliers) OR (better) knowledge of chemical purity / chemical quality OR lower transportation / travel costs (between sites) OR prevents (more) chemical waste through transfer losses OR energy produced in exothermic reactions can be used (in endothermic processes) OR smaller workforce required OR less land required OR saves time so cheaper operational costs.

❖ Catalysts and Catalytic Converters:

- The catalytic converter contains metal oxides. When the exhaust gases pass through the catalytic converter, ammonia reacts with NO_x gases to form nitrogen and water. Give the reason why it is **not** correct to state that urea is acting as a catalyst in the reaction.
 - the urea is used up / is not reformed (in the reaction)
- ALLOW urea mass reduces / amount reduces / reacts to form other products OR

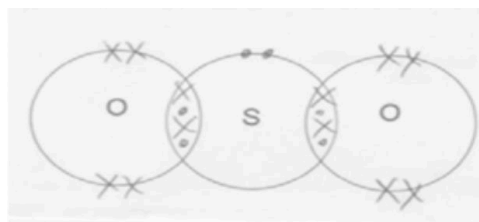
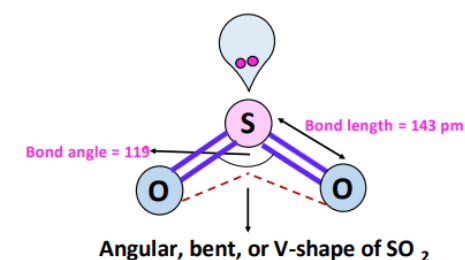
"catalysts are not used up in the reaction"

- The catalytic converter works best at a temperature of around 400°C. Suggest how the catalytic converter reaches this temperature.
 - The hot gases from the engine warm up the catalytic converter*
Allow warmed up by heat produced from combustion.
- Catalysts such as phosphoric acid are bonded to a support material that contains. Suggest the advantage of using support materials containing lots of pores.
 - large(r) surface area (so faster rate of reaction).*

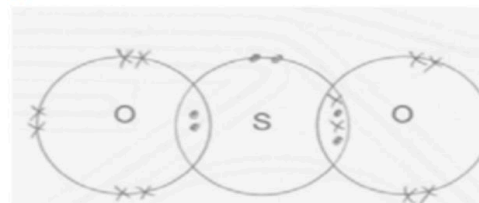
❖ *The Environment:*

- (R.Q.): Give two reasons why it is important to remove nitrogen oxides from the exhaust gases of diesel engines.
 - oxides of nitrogen dissolve in water to form acid rain/nitric acid*
 - oxides of nitrogen cause breathing problems /asthma*
 - depletion of the ozone layer*
 - formation of (photochemical) smog*
- Ozone depletion:

$$2O_3 \rightarrow 3O_2$$
- Give two reasons why the presence of a small number of chlorine free radicals in the upper atmosphere causes a large decrease in the amount of ozone.
 - many ozone molecules decompose for each free radical formed / a single free radical breaks down a large number of ozone molecules.*
 - the chlorine free radical is regenerated / recycled / does not get used up (in the reaction).*
 - chlorine free radicals cause a chain reaction / reaction can be repeated many times / reaction produces another free radical.*
- Sulfur dioxide:



or

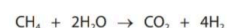


- Droplets of sulfuric acid are formed when sulfur trioxide reacts with water in the upper atmosphere. These droplets reflect ultraviolet radiation from the Sun back into space.
- In the volcanic eruption in Indonesia in 1815, large amounts of both carbon dioxide and sulfur dioxide were released. During the following year, there were global low temperatures. Explain the relative effect of both gases on global temperatures after the volcanic eruption.
 - carbon dioxide causes a rise in temperature*
 - (sulfur dioxide results in the formation of) sulfuric acid which causes global cooling / causes a decrease in temperature*
 - the effect of sulfur dioxide is greater than that of carbon dioxide (because the global temperature decreased)*

Hydrogen can be obtained by two methods.

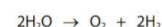
Method 1

The hydrogen is usually obtained by reacting methane gas with steam.



Method 2

Hydrogen can also be obtained using solar power to split water into hydrogen and oxygen.



- Evaluate which of these two methods used to obtain hydrogen is more sustainable.

- *method 2 is more sustainable because it does not use methane/fossil fuels*
- *methane/fossil fuels are a finite resource / solar power is renewable*
- *because it does not produce CO₂ / does not produce greenhouse gases / contribute to global warming.*
- Methanol produced from petrol by CCU may be mixed with petrol derived from fossil fuels. A concentration of 5% by mass of methanol is used. Explain how the use of this fuel would affect the increase in global temperatures.
 - *burning (petrol/methanol/fuel) produces CO₂ /greenhouse gases/contributes to global warming*
 - *the methanol is made from captured CO₂ / is made from CO₂ that would have been released anyway*
 - *the increase in global temperatures will be reduced.*

❖ Bioalcohols and Fermentation:

- Give one advantage and one disadvantage of using bioethanol in petrol.
 - *advantage: (it is produced from) renewable / sustainable (resources) OR it produces fewer emissions (of CO₂ / greenhouse gases)*
 - *disadvantage: it uses up land (that could be used for food production)/uses a food crop that could be eaten OR it produces less energy (per mole / g).*
- Suggest why fermentation of alcohol (ethanol) must be carried out in the absence of air.
 - *ethanol can be oxidised into carboxylic acid / ethanoic acid / aldehyde / ethanal ALLOW ethanol can react with oxygen/ to ensure only anaerobic respiration takes place/ aerobic respiration would take place (in the air) (producing water and carbon dioxide).*
- Suggest how the ethanol can be obtained, after filtering the fermentation mixture.
 - *(fractional) distillation / distill the mixture.*
- Suggest a problem arising from the hygroscopic (ability to absorb moisture) nature of ethanol when using this fuel in a motor vehicle.
 - *(water can cause) corrosion / rusting*
 - *reduce the energy efficiency of the*

engine/fuel/ethanol

- *causes phase separation of the fuel.*

❖ Reactions and Alloys:

- Write the ionic equation for the reaction of silver(I) oxide with water. State symbols are not required. And suggest one advantage of using silver(I) oxide rather than sodium hydroxide or potassium hydroxide in reaction of halogenoalkanes with OH⁻ ions.
 - *ionic eqn: $\text{Ag}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{Ag}^+ + 2\text{OH}^-$*
 - *Advantage: (because silver(I) oxide is insoluble) it can easily be separated from the reaction mixture (when reaction is complete).*
- Titanium(IV) chloride is a liquid that reacts violently with water, producing white smoke. This reaction has been used to produce naval smoke screens to hide ships. Suggest the type of reaction taking place and the compound producing the white smoke.
 - *type: hydrolysis*
 - *compound: titanium(IV) oxide/TiO₂*
- Suggest a reason why this alloy (Li and Al) is better than pure aluminium for aircraft manufacture.
 - *makes the alloy stronger OR (being less dense than Al, the Li would make) the alloy less dense(so would make the aircraft lighter so less fuel used)*